Comparasion and Reflection of Pricing Method

P/E Ratio

- P/E=price/earning per share (EPS)
- Price
  - normally it is current price
  - someone also use the average price of the past 6 months
- EPS
  - time varies
    - » recent
    - » past four seasons
    - » the coming one year

Understand P/E Ratio

- Why do P/E ratios differ from nations, industries, companies?
- What factors decide P/E ratio?
- How to know whether the current P/E ratio sustains?

Back to DCF Model——Gordon Growth Model

- Gordon model, also known as constant growth model, which assumes dividends grow at a constant rate g:
  - （r 〉 g）

$$P_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \ldots = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

$$\frac{P_0}{E_0} = \frac{\frac{D_0}{E_0}(1+g)}{r-g} = \frac{z(1+g)}{r-g}$$

Life Cycle and Growth Rate at Different Stages

TABLE 18.2

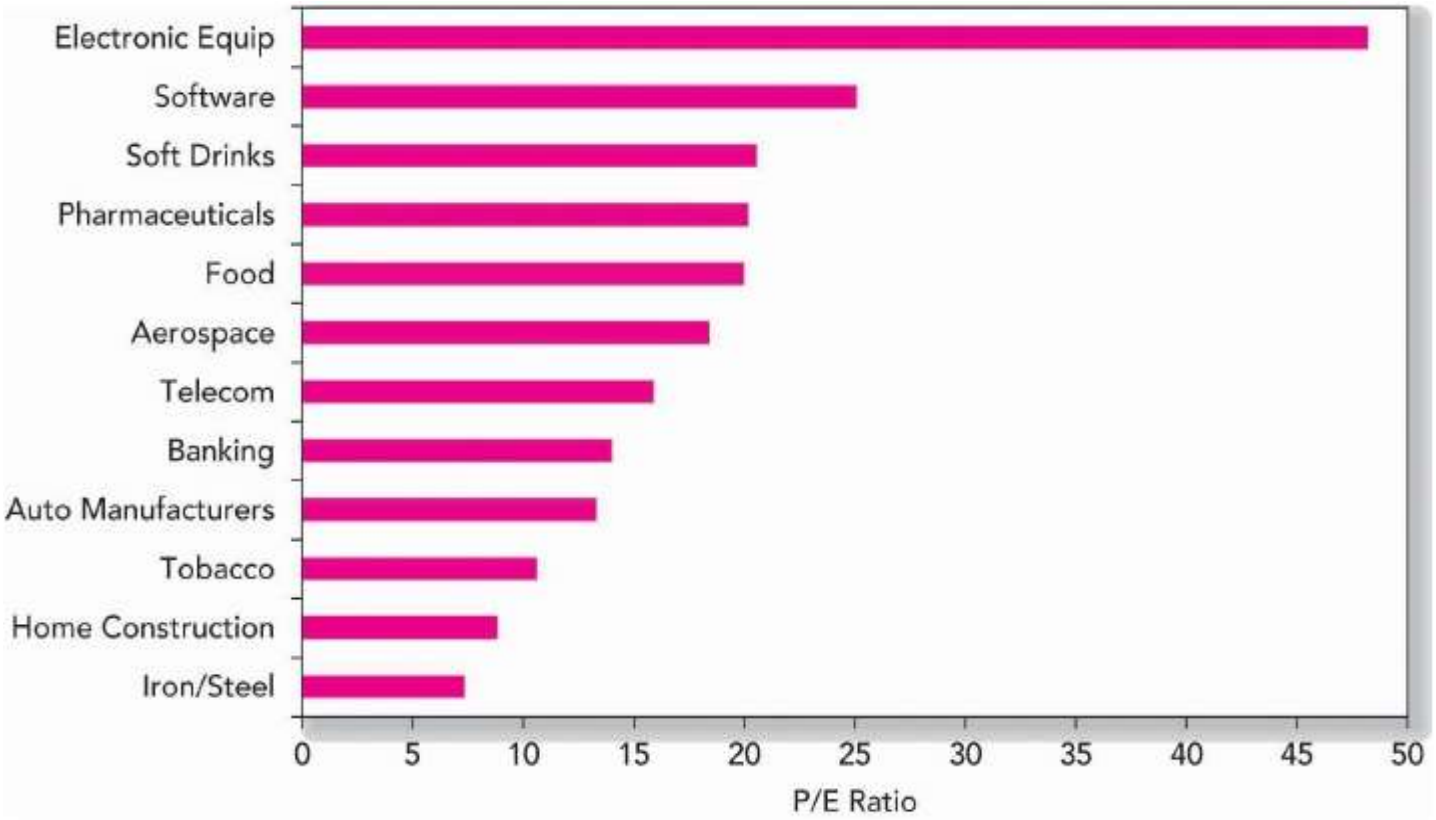
Financial ratios in two industries

	Return on Assets	Payout Ratio	Growth Rate 2005–2008
Computer Software			
Adobe Systems	21.5%	1.0%	8.2%
Cognizant	19.0	0.0	22.8
Compuware	10.5	0.0	17.6
Intuit	19.0	0.0	8.0
Microsoft	31.5	35.0	15.4
Novell	8.5	0.0	51.8
Oracle	33.0	0.0	18.6
Red Hat	17.0	0.0	17.6
Parametric Tech	20.0	0.0	33.9
SAP	22.5	18.0	13.8
Median	19.5%	0.0	17.6%
Electric Utilities			
Central Hudson G&E	6.0%	78.0%	5.1%
Central Vermont	7.5	60.0	8.0
Consolidated Edison	5.0	75.0	1.0
Duquesne Light	8.0	85.0	7.7
Energy East	6.0	74.0	4.1
Northeast Utilities	5.0	59.0	14.0
Nstar	8.5	61.0	3.2
Pennsylvania Power	11.0	52.0	9.3
Public Services Enter.	7.0	62.0	1.7
United Illuminating	5.0	113.0	1.3
Median	6.5%	68.0%	4.6%

Source: *Value Line Investment Survey*, 2006. Reprinted with permission of Value Line Investment Survey. © 2006 Value Line Publishing, Inc. All rights reserved.

Growth Opportunity and P/E Ratio

- the more growth opportunities, the higher P/E



P/E Ratio

- From the definition, P/E ratio is affected by
  - growth rate
  - discounted rate
  - dividend payout ratio (or reinvestment ratio)

$$P_0 = \frac{D_0(1+g)}{(1+r)} + \frac{D_0(1+g)^2}{(1+r)^2} + \ldots = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

$$\frac{P_0}{E_0} = \frac{\frac{D_0}{E_0}(1+g)}{r-g} = \frac{z(1+g)}{r-g}$$

What kind of Chinese companies have high P/E ratio?

- PE=24.08-7.50*beta+2.62*payratio+6.95*exp_5y_growth_in_EPS-1.19*std
  （7.23） （2.58） （3.43） （0.45）

  R-square：20%

- Conclusion： the Chinese companies with high P/E
  - low beta
  - relatively high dividend payout ratio
  - high expected growth rate on profit

  - but have no close relationship with
    - » volatility of stocks

Analysis of Target

- Hard to find appropriate targets sometimes
- Instead of looking for appropriate targets, try to study the relationship between P/E ratio and other factors
- Regression is also a way

American Beverage Company

Company Name	Trailing PE	Expected Growth	Standard Deviation
Coca-Cola Bottling	29.18	9.50%	20.58%
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%
Anheuser-Busch	24.31	11.00%	22.92%
Corby Distilleries Ltd.	16.24	7.50%	23.66%
Chalone Wine Group Ltd.	21.76	14.00%	24.08%
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%
Todhunter Int'l.	8.94	3.00%	25.74%
Brown-Forman 'B'	10.07	11.50%	29.43%
Coors (Adolph) 'B'	23.02	10.00%	29.52%
PepsiCo, Inc.	33.00	10.50%	31.35%
Coca-Cola	44.33	19.00%	35.51%
Boston Beer 'A'	10.59	17.13%	39.58%
Whitman Corp.	25.19	11.50%	44.26%
Mondavi (Robert) 'A'	16.47	14.00%	45.84%
Coca-Cola Enterprises	37.14	27.00%	51.34%
Hansen Natural Corp.	9.70	17.00%	62.45%

Source: Value Line Database.

Expected growth: the expected growth rate in the next five years

Standard deviation: volatility of stocks

Is Coca-Cola Expensive?

$$PE = 20.87 - 63.98\ \text{Standard deviation} + 183.24\ \text{Expected growth} \qquad R^2 = 51\%$$
$$[3.01] \quad [2.63] \qquad\qquad [3.66]$$

$$\text{Predicted } PE_{\text{Coca-Cola}} = 20.87 - 63.98(.3551) + 183.24(.19) = 32.97$$

What can explain the P/E ratio of a country?

Country	*PE Ratio*	*Interest Rates*	*GDP Real Growth*	*Country Risk*
Argentina	14	18.00%	2.50%	45
Brazil	21	14.00%	4.80%	35
Chile	25	9.50%	5.50%	15
Hong Kong	20	8.00%	6.00%	15
India	17	11.48%	4.20%	25
Indonesia	15	21.00%	4.00%	50
Malaysia	14	5.67%	3.00%	40
Mexico	19	11.50%	5.50%	30
Pakistan	14	19.00%	3.00%	45
Peru	15	18.00%	4.90%	50
Phillipines	15	17.00%	3.80%	45
Singapore	24	6.50%	5.20%	5
South Korea	21	10.00%	4.80%	25
Thailand	21	12.75%	5.50%	25
Turkey	12	25.00%	2.00%	35
Venezuela	20	15.00%	3.50%	45

What can explain the P/E ratio of a country?

- PE = 16.16 - 7.94 Interest Rates + 154.40 Growth in GDP - 0.1116 Country Risk

  R Squared = 73%

- Conclusion
  - a country would have a high P/E if
    - » it has low interest rates
    - » it has high GDP growth rate
    - » it has low risks

What can explain the P/E ratio of a country?

Country	*PE Ratio*	*Interest Rates*	*GDP Real Growth*	*Country Risk*	*Predicted PE*
Argentina	14	18.00%	2.50%	45	13.57
Brazil	21	14.00%	4.80%	35	18.55
Chile	25	9.50%	5.50%	15	22.22
Hong Kong	20	8.00%	6.00%	15	23.11
India	17	11.48%	4.20%	25	18.94
Indonesia	15	21.00%	4.00%	50	15.09
Malaysia	14	5.67%	3.00%	40	15.87
Mexico	19	11.50%	5.50%	30	20.39
Pakistan	14	19.00%	3.00%	45	14.26
Peru	15	18.00%	4.90%	50	16.71
Phillipines	15	17.00%	3.80%	45	15.65
Singapore	24	6.50%	5.20%	5	23.11
South Korea	21	10.00%	4.80%	25	19.98
Thailand	21	12.75%	5.50%	25	20.85
Turkey	12	25.00%	2.00%	35	13.35
Venezuela	20	15.00%	3.50%	45	15.35

PEG

- Is a company having a high P/E ratio due to its high price or high growth?

- The definition of PEG

  = P/E ratio / expected growth rate

Which one is more expensive, Pepsi or Coca-Cola?

Company Name	Trailing PE	Expected Growth	Standard Deviation
Coca-Cola Bottling	29.18	9.50%	20.58%
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%
Anheuser-Busch	24.31	11.00%	22.92%
Corby Distilleries Ltd.	16.24	7.50%	23.66%
Chalone Wine Group Ltd.	21.76	14.00%	24.08%
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%
Todhunter Int'l.	8.94	3.00%	25.74%
Brown-Forman 'B'	10.07	11.50%	29.43%
Coors (Adolph) 'B'	23.02	10.00%	29.52%
PepsiCo, Inc.	33.00	10.50%	31.35%
Coca-Cola	44.33	19.00%	35.51%
Boston Beer 'A'	10.59	17.13%	39.58%
Whitman Corp.	25.19	11.50%	44.26%
Mondavi (Robert) 'A'	16.47	14.00%	45.84%
Coca-Cola Enterprises	37.14	27.00%	51.34%
Hansen Natural Corp.	9.70	17.00%	62.45%

Which one is more expensive, Pepsi or Coca-Cola?

The PE ratios and expected growth rates in EPS over the next five years, based on consensus estimates from analysts, for the firms that are categorized as beverage firms are summarized in the following table:

Company Name	Trailing PE	Expected Growth	Standard Deviation	PEG
Coca-Cola Bottling	29.18	9.50%	20.58%	3.07
Molson Inc. Ltd. 'A'	43.65	15.50%	21.88%	2.82
Anheuser-Busch	24.31	11.00%	22.92%	2.21
Corby Distilleries Ltd.	16.24	7.50%	23.66%	2.16
Chalone Wine Group Ltd.	21.76	14.00%	24.08%	1.55
Andres Wines Ltd. 'A'	8.96	3.50%	24.70%	2.56
Todhunter Int'l.	8.94	3.00%	25.74%	2.98
Brown-Forman 'B'	10.07	11.50%	29.43%	0.88
Coors (Adolph) 'B'	23.02	10.00%	29.52%	2.30
PepsiCo, Inc.	33.00	10.50%	31.35%	3.14
Coca-Cola	44.33	19.00%	35.51%	2.33
Boston Beer 'A'	10.59	17.13%	39.58%	0.62
Whitman Corp.	25.19	11.50%	44.26%	2.19
Mondavi (Robert) 'A'	16.47	14.00%	45.84%	1.18
Coca-Cola Enterprises	37.14	27.00%	51.34%	1.38
Hansen Natural Corp.	9.70	17.00%	62.45%	0.57
Average	22.66	12.60%	33.30%	2.00

P/B Ratio

- PB=price/book value of asset per share
  - Pros
  - Cons

Factors Affecting P/B Ratio

$$P_0 = \frac{D_1}{(k - g_n)}$$

$$\frac{P}{BV} = \frac{ROE \times z}{k - g_n}$$

$$\frac{P}{BV} = \frac{ROE - g_n}{k - g_n}$$

k = discounted rate； z = dividend payout ratio； P_0 = price；

g_n = the growth rate of earning and dividends

D_1 = dividend next year, BV=book value

Select Stocks through P/B

The following table reports on the price-to-book ratios for integrated oil companies listed in th United States in September 2000:

Company Name	*Ticker Symbol*	*Price-to-Book Ratio*	*Return on Equity*	*Standard Deviation*
Crown Central Petroleum "A"	CNPA	0.29	–14.60%	59.36%
Giant Industries	GI	0.54	7.47%	38.87%
Harken Energy Corp.	HEC	0.64	–5.83%	56.51%
Getty Petroleum Mktg.	GPM	0.95	6.26%	58.34%
Pennzoil–Quaker State	PZL	0.95	3.99%	51.06%
Ashland Inc.	ASH	1.13	10.27%	21.77%
Shell Transport	SC	1.45	13.41%	31.61%
USX–Marathon Group	MRO	1.59	13.42%	45.31%
Lakehead Pipe Line	LHP	1.72	13.28%	19.56%
Amerada Hess	AHC	1.77	16.69%	26.89%
Tosco Corp.	TOS	1.95	15.44%	34.51%
Occidental Petroleum	OXY	2.15	16.68%	39.47%
Royal Dutch Petroleum	RD	2.33	13.41%	29.81%
Murphy Oil Corp.	MUR	2.40	14.49%	27.80%
Texaco Inc.	TX	2.44	13.77%	27.78%
Phillips Petroleum	P	2.64	17.92%	29.51%
Chevron Corp.	CHV	3.03	15.69%	26.44%
Repsol-YPF ADR	REP	3.24	13.43%	26.82%
Unocal Corp.	UCL	3.53	10.67%	34.90%
Kerr-McGee Corp.	KMG	3.59	28.88%	42.47%
Exxon Mobil Corp.	XOM	4.22	11.20%	19.22%
BP Amoco ADR	BPA	4.66	14.34%	27.00%
Clayton Williams Energy	CWEI	5.57	31.02%	26.31%
Average		2.30	12.23%	

Select Stocks through P/B

$$PBV = 2.21 + 8.22\ ROE - 2.63\ Standard\ deviation \qquad R^2 = 52\%$$
$$[2.16]\ [2.92] \qquad [1.21]$$

Company Name	*Price-to-Book Ratio*	*Predicted PBV*	*Under- or Overvalued*
Crown Central Petroleum "A"	0.29	–0.56	NMF
Giant Industries	0.54	1.80	–69.74%
Harken Energy Corp.	0.64	0.24	166.59%
Getty Petroleum Mktg.	0.95	1.19	–19.67%
Pennzoil–Quaker State	0.95	1.19	–19.93%
Ashland Inc.	1.13	2.48	–54.28%
Shell Transport	1.45	2.48	–41.56%
USX–Marathon Group	1.59	2.12	–25.11%
Lakehead Pipe Line	1.72	2.78	–38.03%
Amerada Hess	1.77	2.87	–38.33%

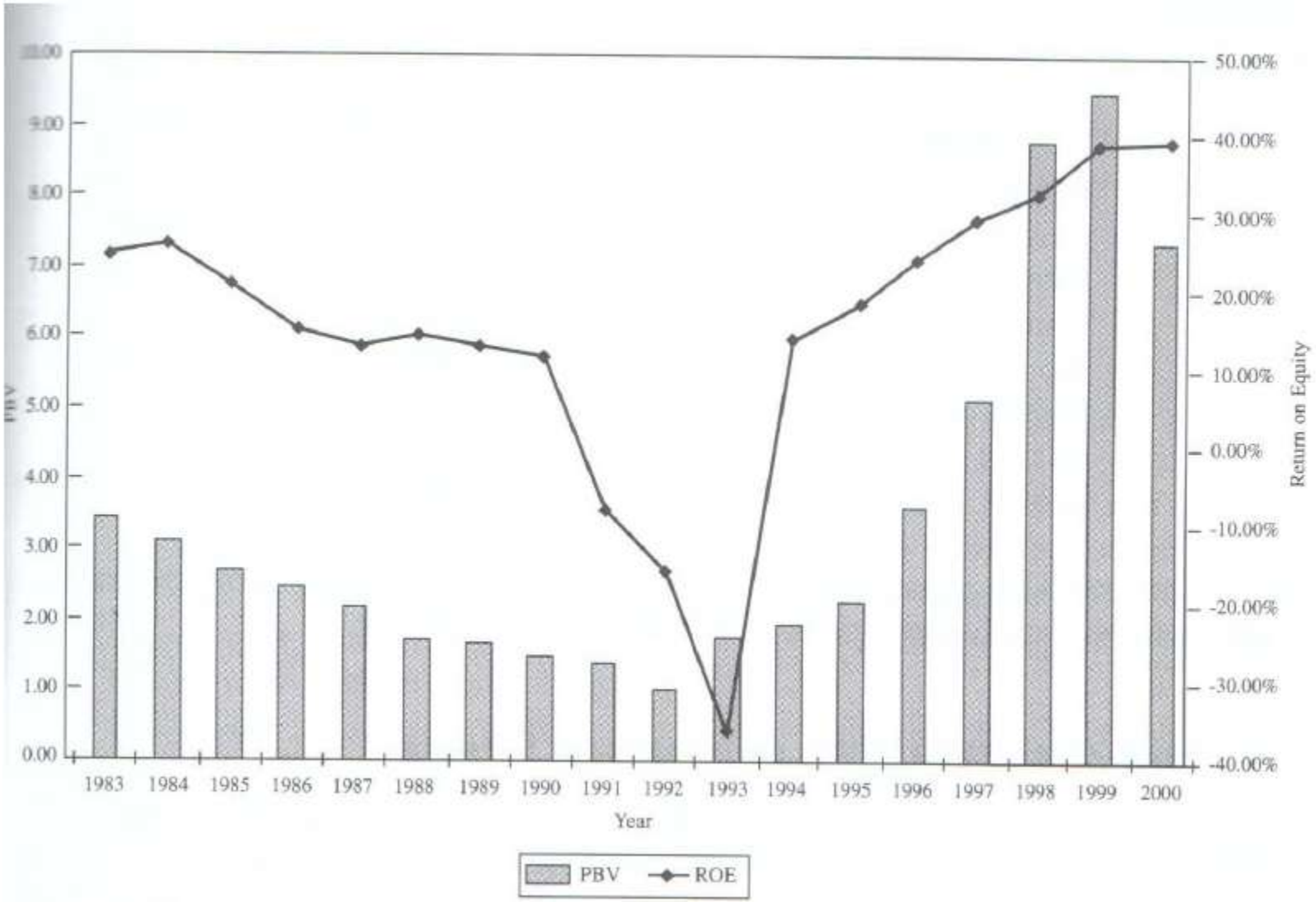
Select Stocks through P/B

Company Name	Price-to-Book Ratio	Predicted PBV	Under- or Overvalued
Tosco Corp.	1.95	2.57	–24.09%
Occidental Petroleum	2.15	2.54	–15.27%
Royal Dutch Petroleum	2.33	2.52	–7.66%
Murphy Oil Corp.	2.40	2.67	–10.07%
Texaco Inc.	2.44	2.61	–6.47%
Phillips Petroleum	2.64	2.90	–9.17%
Chevron Corp.	3.03	2.80	8.20%
Repsol-YPF ADR	3.24	2.60	24.53%
Unocal Corp.	3.53	2.17	63.05%
Kerr-McGee Corp.	3.59	3.46	3.70%
Exxon Mobil Corp.	4.22	2.62	60.99%
BP Amoco ADR	4.66	2.67	74.03%
Clayton Williams Energy	5.57	4.06	36.92%

Which one is the most expensive?

Which one is the cheapest?

The P/B and Return of IBM



Price/Sales Ratio

- Pros
  - less likely to be manipulated, compared to earnings and book value
  - earnings may be negative or close to zero
  - more stable than P/E
  - more suitable to price mature, cyclical company with close zero profit

Price/Sales Ratio

- Cons
  - large sales is not equal to high profit
  - stock price should not tightly related to sales

- If there are two stocks having the same P/E, which one would have higher price/sales ratio?

Price/Sales Model： 1-stage Model

$$P_0 = \frac{D_1}{(k - g_n)} \Rightarrow \frac{P}{S} = \frac{PM \times z \times (1 + g_n)}{k - g_n} \qquad \frac{P_0}{E_0} = \frac{\frac{D_0}{E_0}(1+g)}{r-g} = \frac{z(1+g)}{r-g}$$

k = discounted rate； z = dividend payout ratio； P_0 = price；

g_n = the growth rate of earning and dividends

D_1 = dividend next year： S= sales； PM = net profit margin=E/S

2-Stage Model

$$\frac{P_0}{S_0} = PM \times \left\{ \frac{z_n \times (1+g) \times \left[1 - \frac{(1+g)^n}{(1+k_{hg})^n} \right]}{(k_{hg} - g)} + \frac{z \times (1+g)^n \times (1+g_n)}{(k_{st} - g_n) \times (1+k_{hg})^n} \right\}$$

PM = net profit margin

g = growth rate in the previous n years

k_{hg} = discounted rate during high growth period

k_{st} = discounted rate during stable growth period

z = dividend payout ratio during high growth period

g_n = growth rate after the first n years

z_n = dividend payout after the first n years

Specific Ratio in Different Industries

Commodity Company（oil refining, gold mining companies）

$$\text{unit commodity value} = \frac{\text{Stock value + debt value - cash}}{\text{commodity inventory}}$$

If manufacturer produce products with identical quality：

$$\text{unit product value} = \frac{\text{stock value + debt value - cash}}{\text{product units（or capacity）}}$$

Specific Ratio in Different Industries

Internet company、information provider、power cable company, etc：

$$\text{Unit user value} = \frac{\text{stock value + debt value - cash}}{\text{number of users}}$$

Amazon：

$$\text{unit customer value} = \frac{\text{stock value + debt value- cash}}{\text{number of effective customers}}$$

$$\text{unit visitor value} = \frac{\text{stock value + debt value - cash}}{\text{number of visitors}}$$

Company with patents: the price when Google bought Motorola

Value of Internet Flow

Company Name	*Firm Value*	*Visitors*	*Value per Visitor*
Lycos, Inc.	$ 5,396.00	5,858	$0.92
MapQuest.com Inc.	$ 604.80	6,621	$0.09
iVillage Inc.	$ 250.40	7,346	$0.03
CNET Networks	$ 1,984.30	10,850	$0.18
Ask Jeeves Inc.	$ 643.50	11,765	$0.05
Go2Net Inc.	$ 1,468.60	12,527	$0.12
LookSmart, Ltd.	$ 1,795.30	13,374	$0.13
About.com Inc.	$ 541.90	18,282	$0.03
Excite@Home	$ 7,008.20	27,115	$0.26
Yahoo! Inc.	$65,633.40	49,045	$1.34

Source: Media Metrix.

Examples of Valuation Application

苏宁

Price (RMB)	10.95
Shares	6,996,211,900
RMB'000	
Equity	76,608,520
Debt	35,348
Cash	21,568,655
EV	55,075,213
Store	962
Area (Sqm)	4,058,300
RMB	
EV/Store	57,250,741
Sales/Store	61,955,525
EV/Sqm	13,571
Sales/Sqm	14,602

国美

Price (RMB)	2.00
Shares	15,055,332,000
RMB'000	
Equity	30,110,664
Debt	3,524,909
Cash	6,029,059
EV	27,606,514
Store	726
Area (Sqm)	2,675,000
RMB	
EV/Store	38,025,501
Sales/Store	58,771,350
EV/Sqm	10,320
Sales/Sqm	15,951

Which one had higher valuation in May 2010?

Examples of Valuation Application

沃尔玛

Price (USD)	50.02
Shares(Million)	3,786
USD Million	
Equity	189,376
Debt	33,754
Cash	7,907
EV	215,223
Store	8,416
Area (Sqm)	
RMB	
EV/Store	174,689,987
Sales/Store	328,677,519

Best Buy

Price (USD)	40.56
Shares(Million)	419
USD Million	
Equity	16,995
Debt	1,802
Cash	1,826
EV	16,971
Store	4,027
Area (Sqm)	
RMB	
EV/Store	28,787,385
Sales/Store	84,283,591

物美

Price (RMB)	16.37
Shares	1,251,274,116
RMB'000	
Equity	20,483,357
Debt	456,086
Cash	1,171,575
EV	19,767,868
Store	472
Area (Sqm)	542,241
RMB	
EV/Store	41,881,077
Sales/Store	24,961,883

Which one is the most expensive?

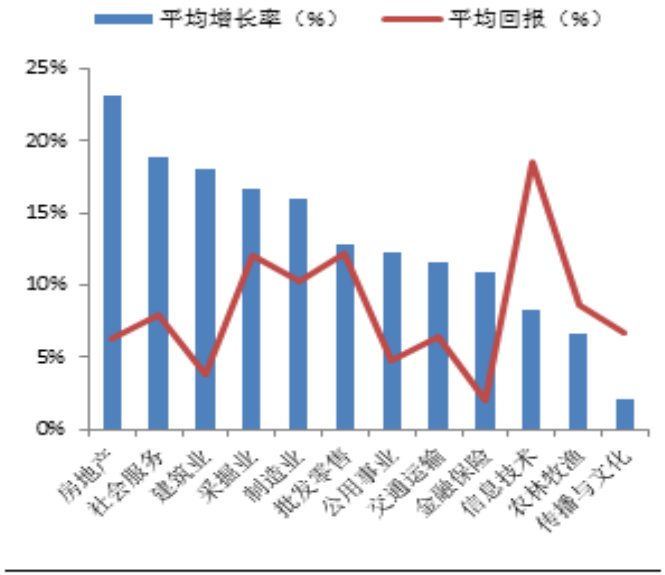
Growth Opportunity and Return

- The growth rate of American computer software companies is 17.6%, while that of electricity companies is only 4.6%. Hence should we buy stocks of computer software companies?

Growth Opportunity and Return

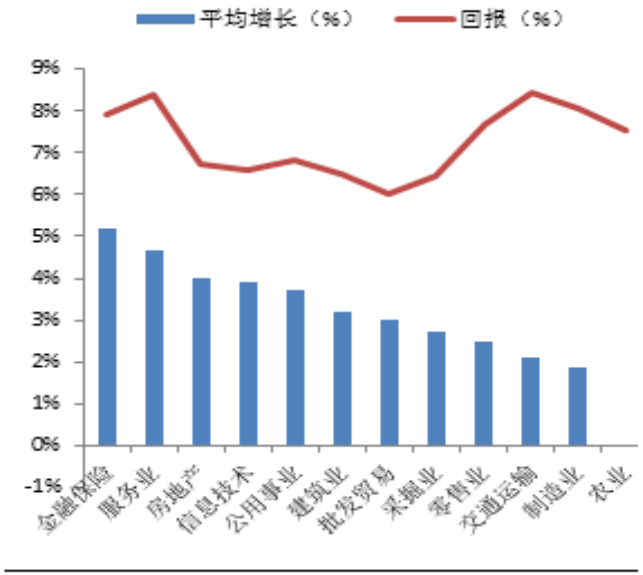
There is no obvious pattern for the growth and return rates in industries of China. The industry with high growth does not mean high return. Take agricultural and real estate industries as examples, the average annual growth rate of real estate is 23.13%, and that of agriculture is 6.69%. If you invest in real estate and agriculture in 1997, you should have returns of 6.27% and 8.58% separately in 2010. The similar things also appear in IT, retail, mining and construction industries. The returns of low growth industry are higher than that of high growth industry.

图 1 中国 1997~2010 各行业增长及年均回报率（增加值）



数据来源：中国增加值数据来源于历年统计年鉴；回报率数值根据 CSMAR 数据库数据计算而得。行业分类采用国民经济行业分类。

图 2 美国 1947~2010 各行业增长及年均回报率（增加值）



说明：农业平均增长率太低，为-0.03%。

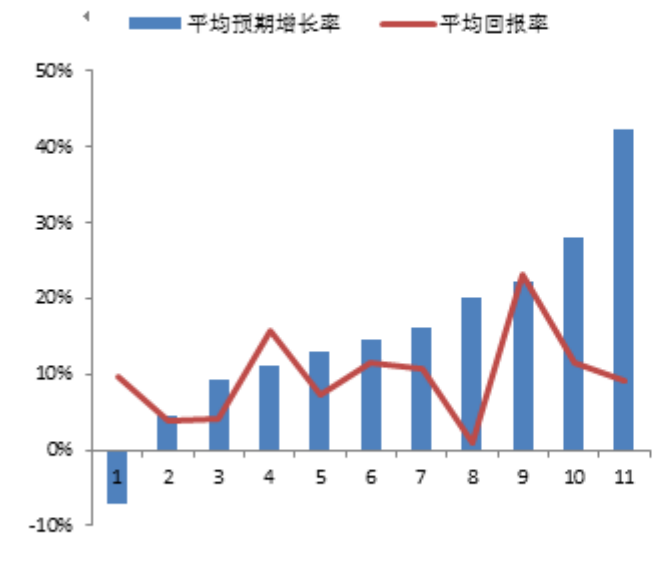
数据来源：美国增加值数据来源于 CEIC 数据库；回报率数值根据 CRSP 数据库数据计算而得；行业分类采用美国 NASIC 分类标准。

Growth Opportunity and Return

Buy the industry with high growth next year/ last year

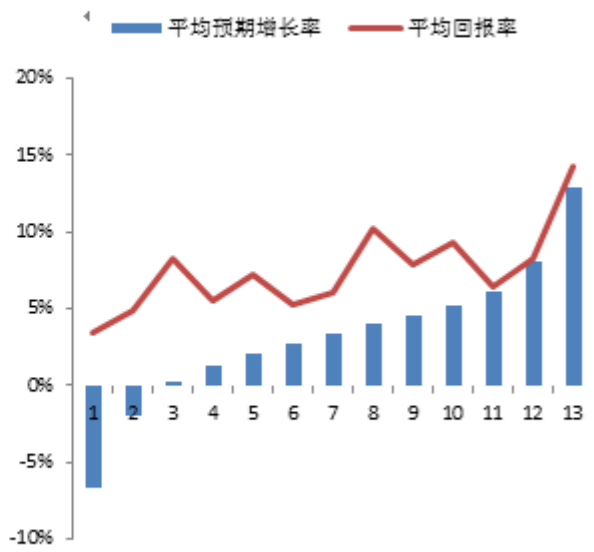
In reality, investors may frequently buy and sell. Assume that investor can estimate the future. They buy at the beginning of each year and sell at the end of year according to the industry growth. Under the assumption of “high growth high return”, stocks in the high growth industries should have higher return.

图 3 中国每年按下一年各行业增长（增加值）情况投资所获收益比较（1997~2010）



注：1~11 分别表示增长由最慢到最快的各组。由于信息技术业增加值不全，剔除该行业。

图 4 美国每年按下一年各行业增长（增加值）情况投资所获收益比较（1947~2010）



注：1~13 分别表示增长由最慢到最快的各组。

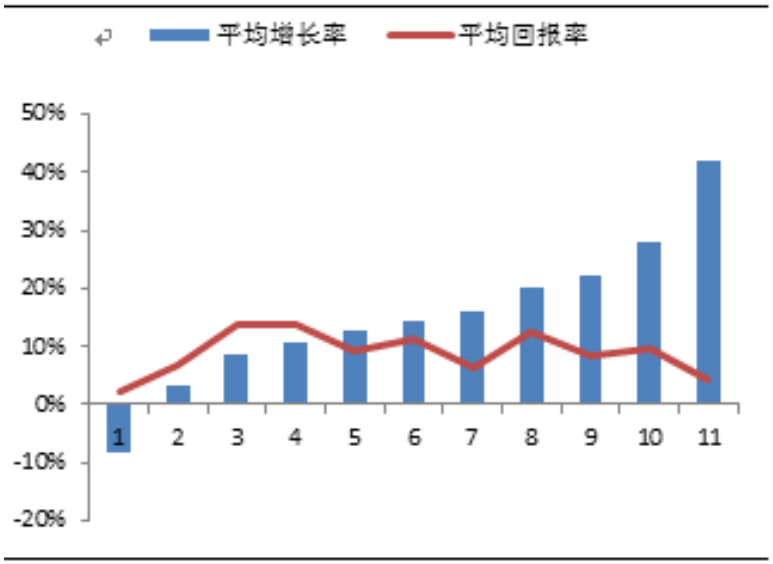
There is no obvious pattern in China, but in the US, it appears that in the industries with the highest and lowest growth rate, there is the pattern.

Jinfan Zhang

Growth Opportunity and Return

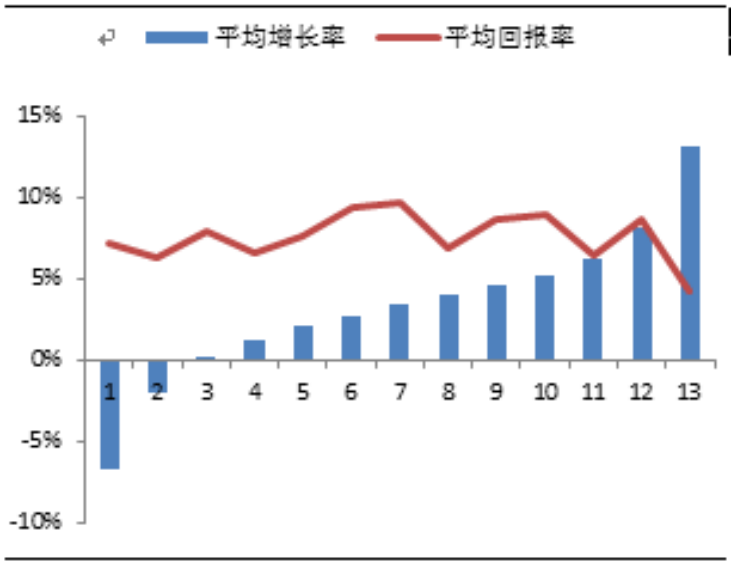
If we take a feasible strategy, which rebalances the portfolio according to the growth rate of the industry last year, no return pattern could be discovered.

图 5 中国每年按上一年各行业增长（增加值）情况投资所获收益比较（1997~2010）



注：1~11 分别表示增长由最慢到最快的各组。由于信息技术业增加值不全，剔除该行业。

图 6 美国每年按上衣年各行业增长（增加值）情况投资所获收益比较（1947~2010）



注：1~13 分别表示增长由最慢到最快的各组。

Worries of Li Guoqing



- The IPO price of Dangdang was $16. The closing price of the first trading day reached $29.91on Dec. 9, 2010.
- In Aug. 2016, Dangdang delisted and the share price was $6.6.

迷失的唯怡：哈哈哈哈哈，此女必火，李国庆你简直就不是人，三个月前好像是你嘴上抹了蜜把一本负盈利的烂帐颤微微地四处磕头求你老婆动用她以前的社会积累和关系拉你一把，做了十年的破网就快倒闭了不想死得这么难看，你别忘了这可是有很多目击证人的。

@当当网李国庆V：摇滚歌词，虚构：为做俺们生意，你们Y给出估值10-60亿，一到香港写招股书，总看韩朝开火，只写7.8亿，别TMD演戏。我大发了脾气。老婆享受辉煌路演，忘了你们为啥窃窃私喜。王八蛋们明知次日开盘就会20亿；还定价16，也就11亿。次日开盘，CFO被股价吓的尿急，我说忍了这口气，过了静默期我再操你妈了逼　原文转发(3952) | 原文评论(1815)

张劲帆

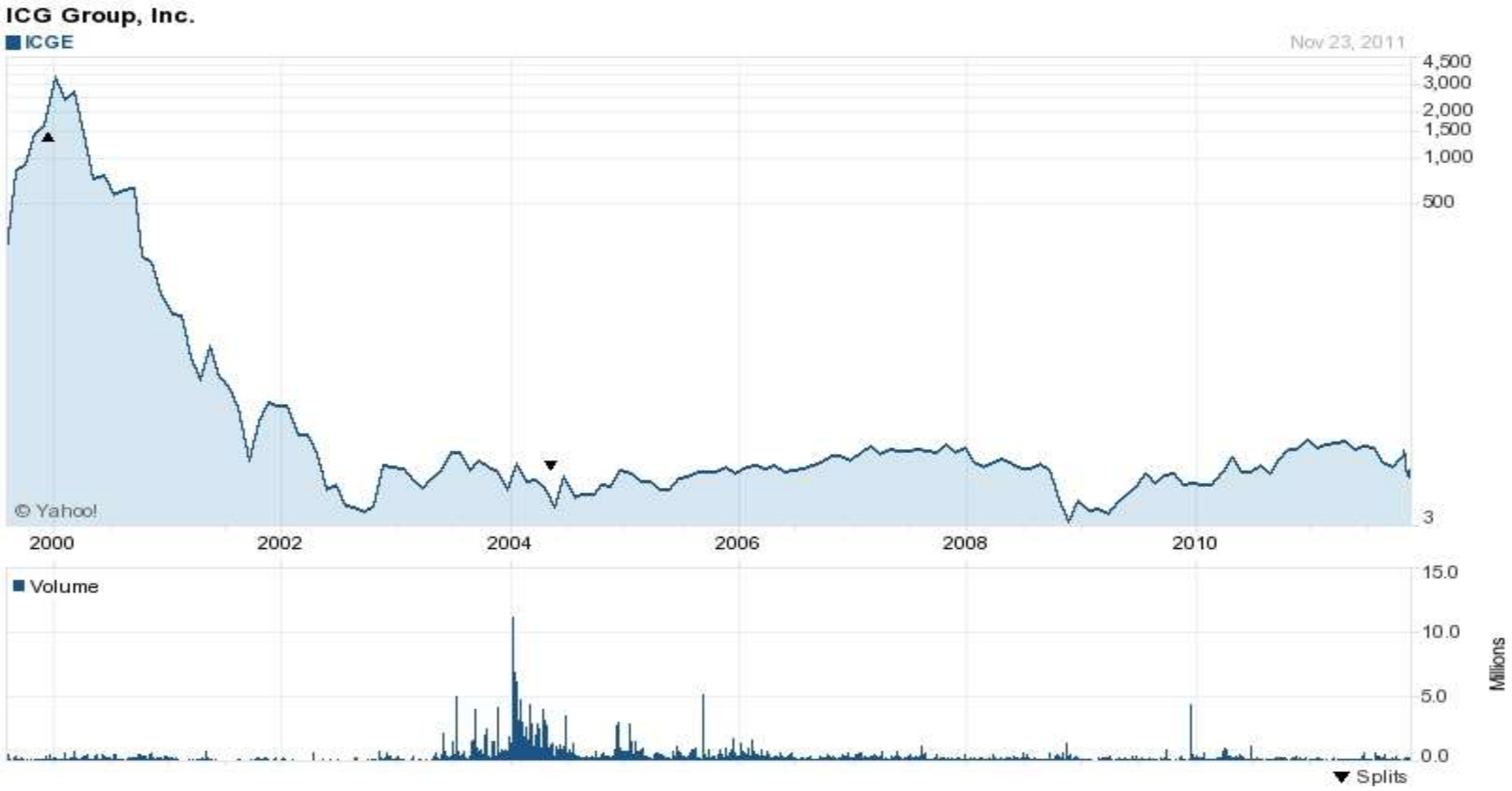
Growth Opportunity and Return

We do not see the solid relationship between high growth and high return. Why?

Market may over expect the growth. The valuation is too high.

Can the stock price be estimated?

The pricing of Internet Capital Group



Nov 23, 2011, $7.52

张劲帆

Looking for Another Fool

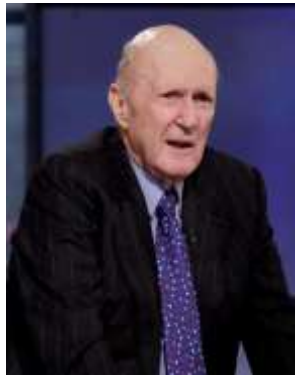
- The pricing of Internet Capital Group
  - "Pricing method is often ineffective when applied to high growth company (Henry Blodget, analyst of Merill Lynch stock said it to Internet Capital Group in Jan 2000.）
- Many investors think that the stock price depend on "supply and demand"
  - supply and demand depend on expectation
  - expectation depends on the "next fool"
- In 1999
  - S&P500 index increased 18%
  - Buffet' Berkshire Hathaway dropped 23%
  - 《Times》： "Warren，what's wrong？"

The NASDAQ index spiked in the late 1990s and then fell sharply as a result of the dot-com bubble.





Warren Buffet

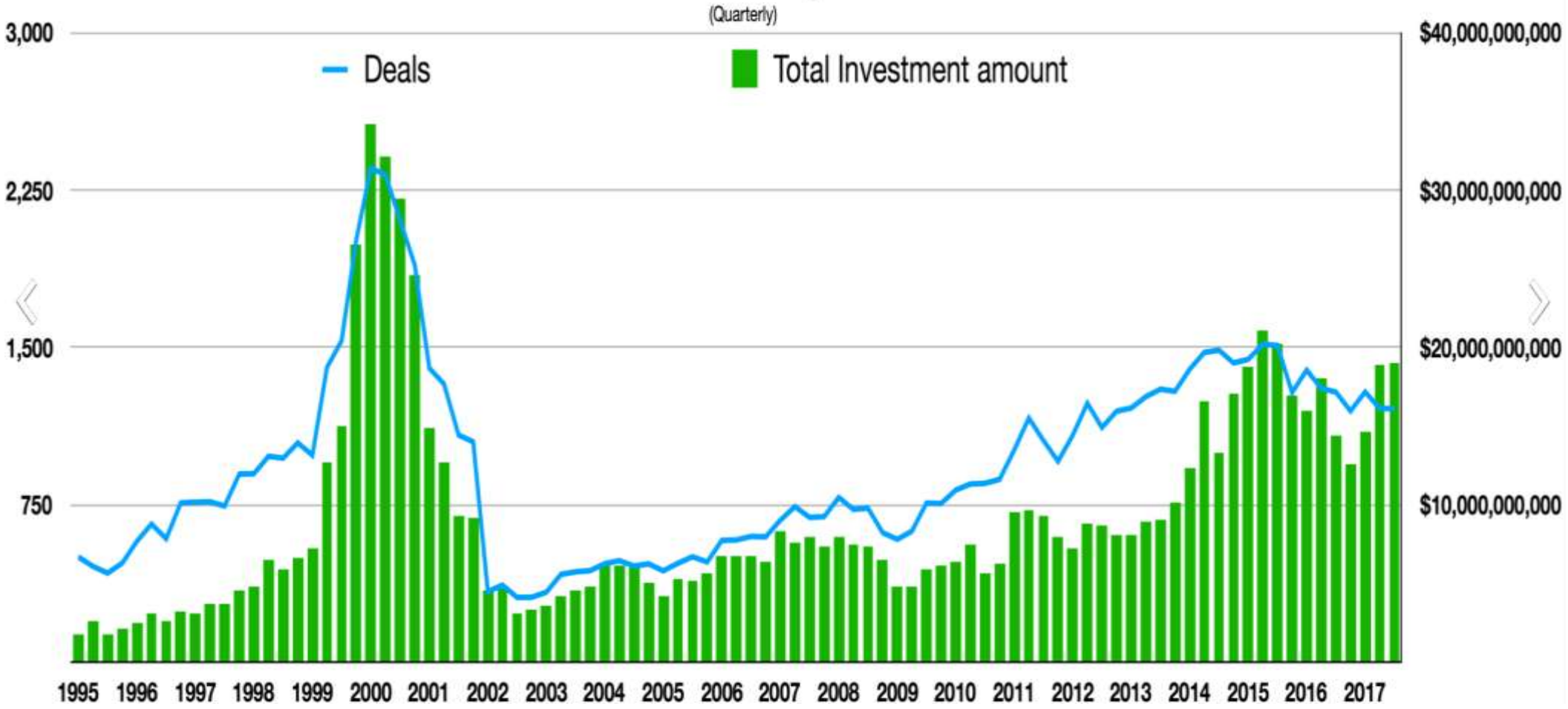


Julian Roberts



George Soros

Total U.S. Venture Capital Investments

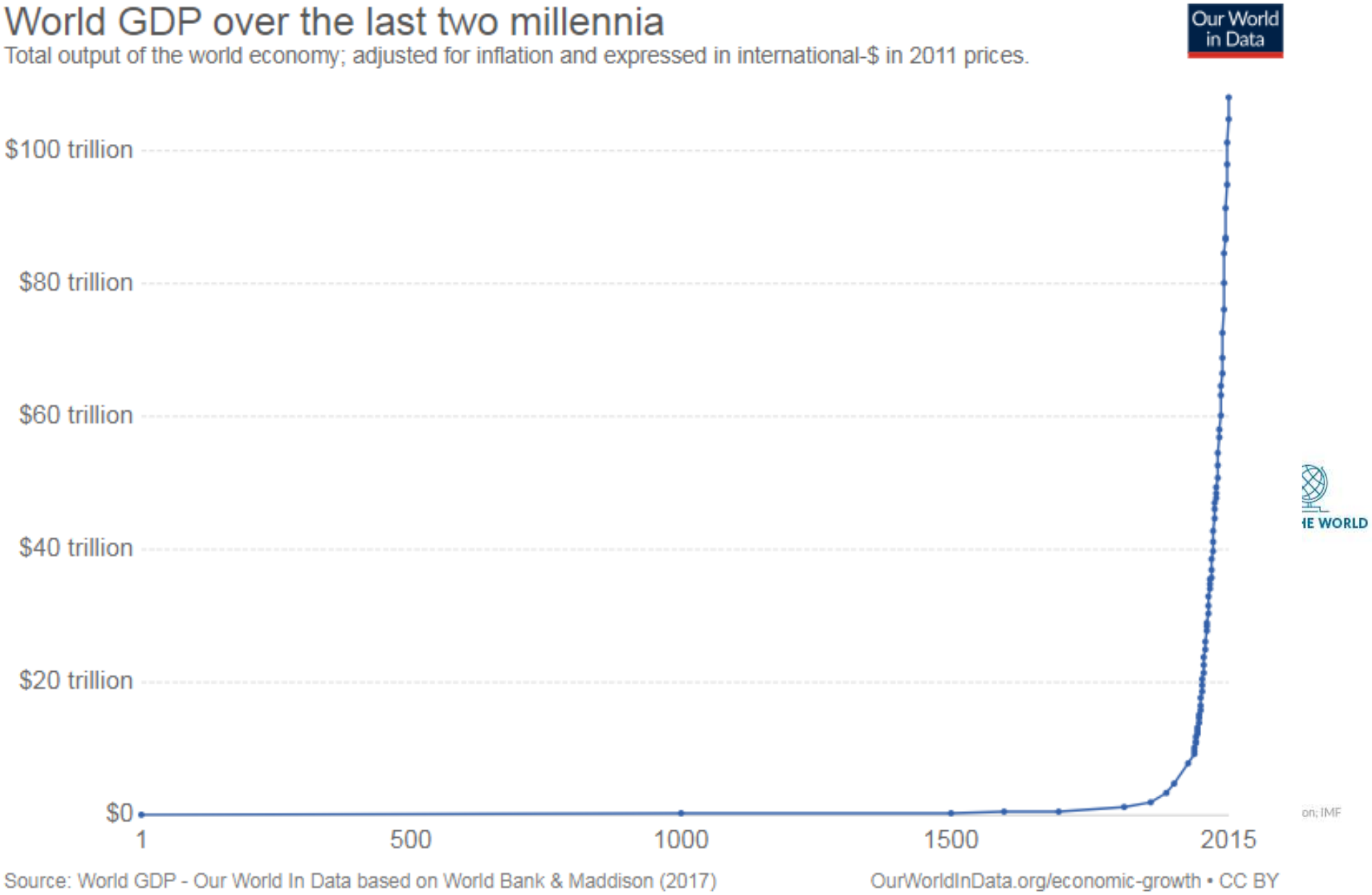




Past 20 years: the peak-to-peak compound return: 1.05^20=2.6=13000/5000
Past 40 years: from 1981-2021 compound return: 1.11^40=65=13000/200

World GDP over the last two millennia

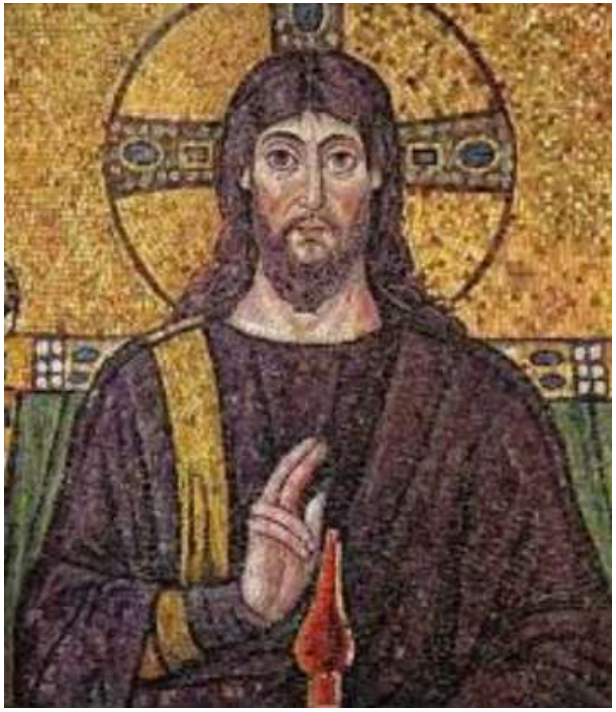
Total output of the world economy; adjusted for inflation and expressed in international-$ in 2011 prices.



Medieval vs Renaissance

humanism









Medieval vs Renaissance







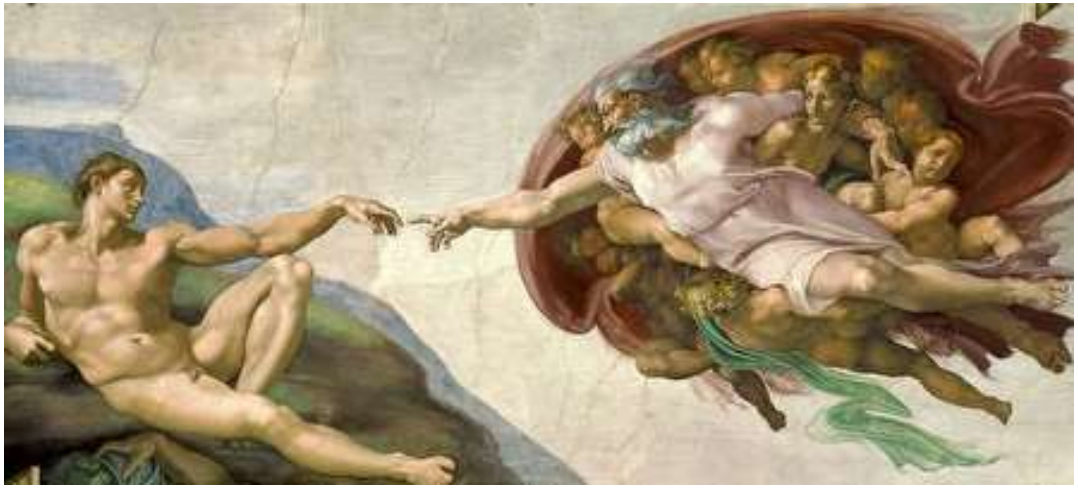
Medieval vs Renaissance

- Sistine Chapel--Michelangelo



Medieval vs Renaissance

Sistine Chapel--Michelangelo

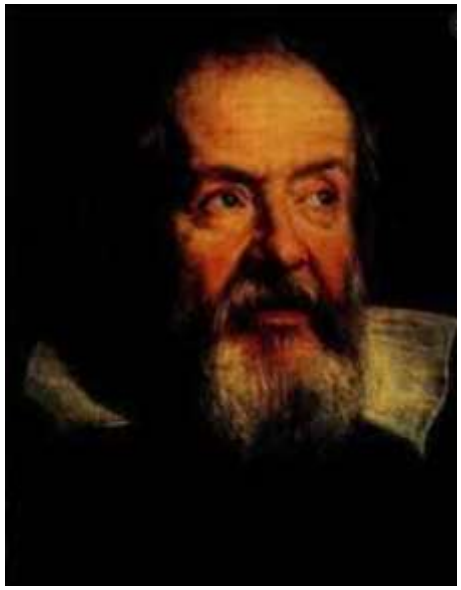






Science

Rationalism



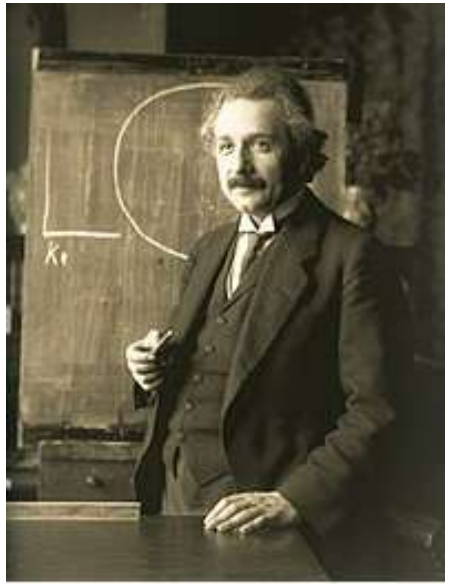
Galileo Galilei
伽利略
(1564－1642)



Isaac Newton
牛顿
(1642－1727)



James Clerk Maxwell
麦克斯韦
(1831－1879）



Albert Einstein
爱因斯坦
(1879－1955)

Can the return of China stock market be forecasted?

Time Series Estimation vs Cross Sectional Estimation



Three short bull markets are separated by long bear markets

1999年5月19日，人民日报发表社论，指出中国股市会有很大发展，投资者踊跃入市，开启了第一波大行情。6月15日人民日报社论《坚定信心、规范发展》给予上涨行情积极评价，助推股市投资热情。

On May 19, 1999, People's Daily published an editorial stating that China's stock market would experience significant development, and investors flocked to the market, initiating the first major wave of the market. On June 15, People's Daily's editorial "Stay Confident, Standardize Development" gave a positive evaluation of the rising market, further boosting investment enthusiasm in the stock market.



2015年6月12日达到5178.19点峰值，创业板市盈率突破140倍。
The peak of 5178.19 points was reached on June 12, 2015, and the P/E ratio of GEM exceeded 140 times.

场内融资规模达2.27万亿元，场外配资约1.8万亿元（杠杆率3-10倍），合计占A股流通市值10%，远超国际警戒线
On-market financing amounted to 2.27 trillion yuan, and over-the-counter funding of about 1.8 trillion yuan (leverage 3-10 times), which together accounted for 10% of the A-share circulating market value, far exceeding the international warning line.

人民日报:内外因共促忐忑行情 十年一遇慢牛已形成

2015年01月30日 09:33 来源：人民日报 参与互动0

查看更多实时行情，点击进入行情中心

今年1月以来，A股市场一改去年流行的"快牛"节奏，在3100点至3400点区间内反复震荡。1月29日，上证综指在多空消息的轮番轰炸下报收于3262.31点，跌幅1.31%，股市未来走向再度成谜。专家指出，股市自身调整需要、投资者情绪起伏、外围市场动荡等都是造成当前行情的主因。未来，随着中国经济"双引擎"的发力，股市有望走出与中国经济发展成就相匹配的"健康牛市"。

人民日报：震荡不改慢牛趋势 短期望冲击4000点

《人民日报海外版》(2015年04月01日 第02版)

周二，A股再次上演"过山车"行情。早盘，沪指高开站上3800点大关，随后在其附近反复拉锯。临近收盘，空方占上风，大盘跳水翻绿。截至收盘，沪指报3747.90点，下跌1.02%；深成指报13160.66点，下跌0.25%，两市成交额近1.3万亿元。面对近期走势，市场各方分歧加剧。专家普遍认为，股市连续上涨本身有调整需求，短期风险释放后仍将随中国经济步入健康的"慢牛"通道。

5月4日，人民日报18版财经纵横刊出《牛市也别忘风险》一文，在连续唱好牛市之后，罕见直接提示股民关注市场风险。这也是本轮牛市以来人民日报第一次在标题上出现"风险"二字。这一评论文章，署名作者午言，近年来人民日报有关股市的评论均出自他手。

人民日报：管理层治理旨在慢牛 长牛势头未变

发布时间：2015-05-08 08:09:15 来源：人民日报 作者：周小苑 责任编辑：王文举

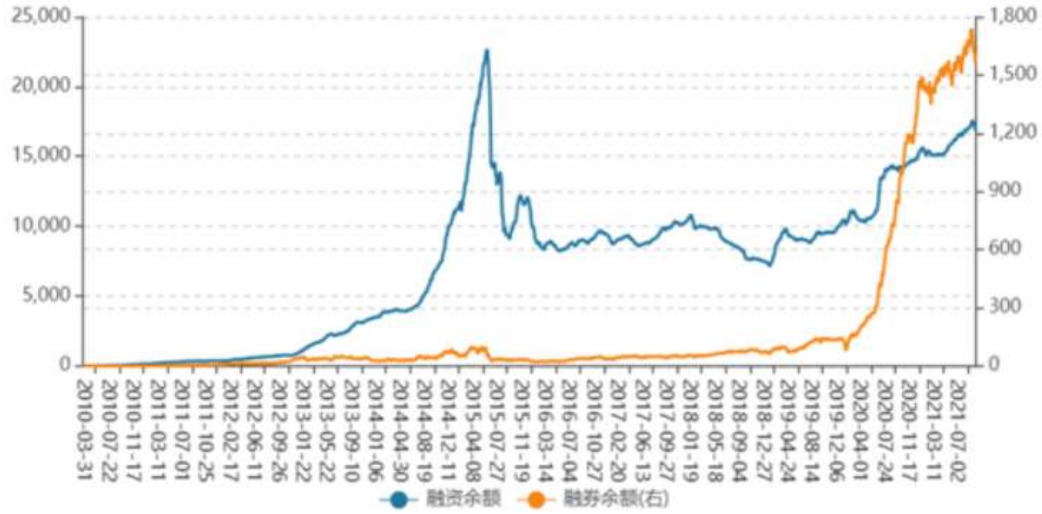


2014年5月下海随即发表：
“论对熊市的最后一战——5000点不是梦”
“党给我智慧给我胆，5000点不是梦”

2015年5月27日，在国泰君安的年中策略会上，任泽平称，“一年前，市场认为我疯了。一年后，我理性了，市场疯了！”。

Phase	Time	SHI(000001) Performance	Main Characteristics
Peak	June 12, 2015	5178.19 pts	Leveraged capital driving the bubble
First Sharp Drop	June 15, 2015	Drops -2%	Panic triggered by the cleanup of margin trading
Market Crash Confirmed	June 26, 2015	Single-Day Drop of -7.04%, Cumulative Drop from June 15 to June 26. Drop from 5178 pts to 4264 pts, a decline of 18%	2000 stocks hit the lower limit in a single trading day
Government Rescue Measures Initiated	June 27: PBC post targeted reduction of required reserve ratio by 0.25% and cut interest rates	Short market rebound followed by continued decline	By July 8, over 1400 listed companies suspend trading to circumvent the market crash, nearly 50% of A-shares
	June 29-30: CSRC issued back-to-back documents to calm the market, and announced a plan for pension funds to enter the market	Limited effect	
	July 4, 2015: 21 securities firms jointly invest 120 billion yuan in blue-chip ETFs, IPO suspension	Limited effect	
Lowest Point	August 26, 2015	2850 pts	45% drop from the peak

图1：两市融资融券余额（亿元）



Margin trading borrowing Short selling borrowing

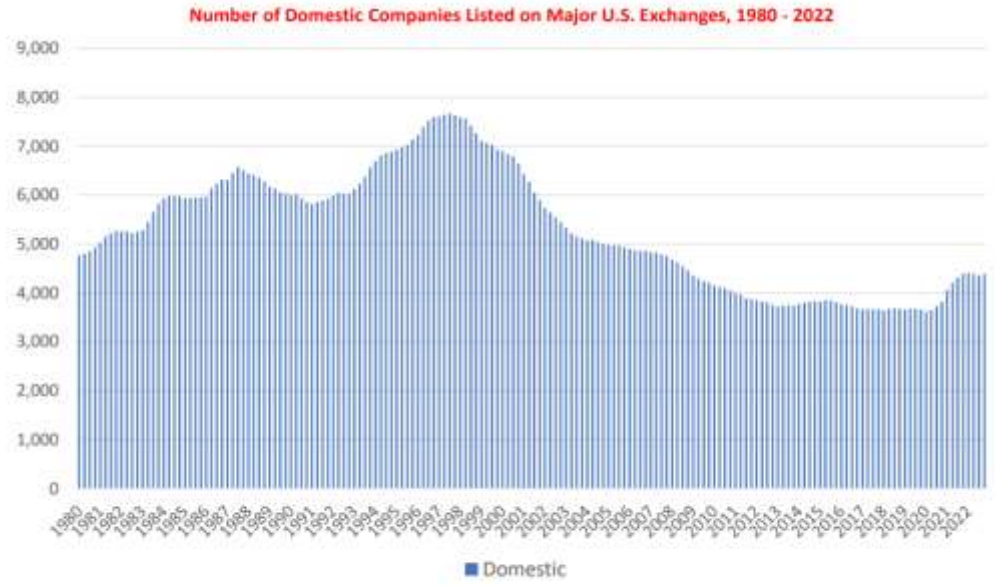
图2：两融余额占A股流通市值、交易额占A股成交额（%）



Margin Trading Balance as a Percentage of A-Share Circulating Market Value, Margin Trading Volume as a Percentage of A-Share Turnover (%)



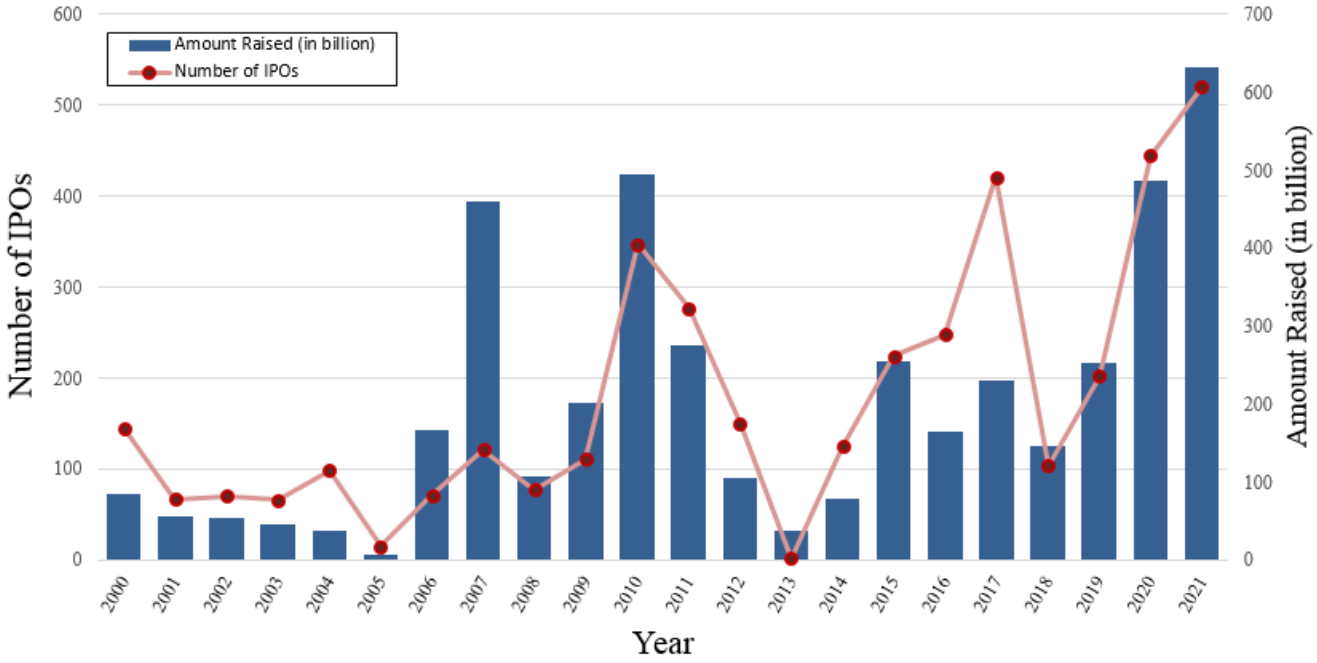
China 2021 GDP 114 Trillion Yuan
Stock Market Cap 90 Trillion



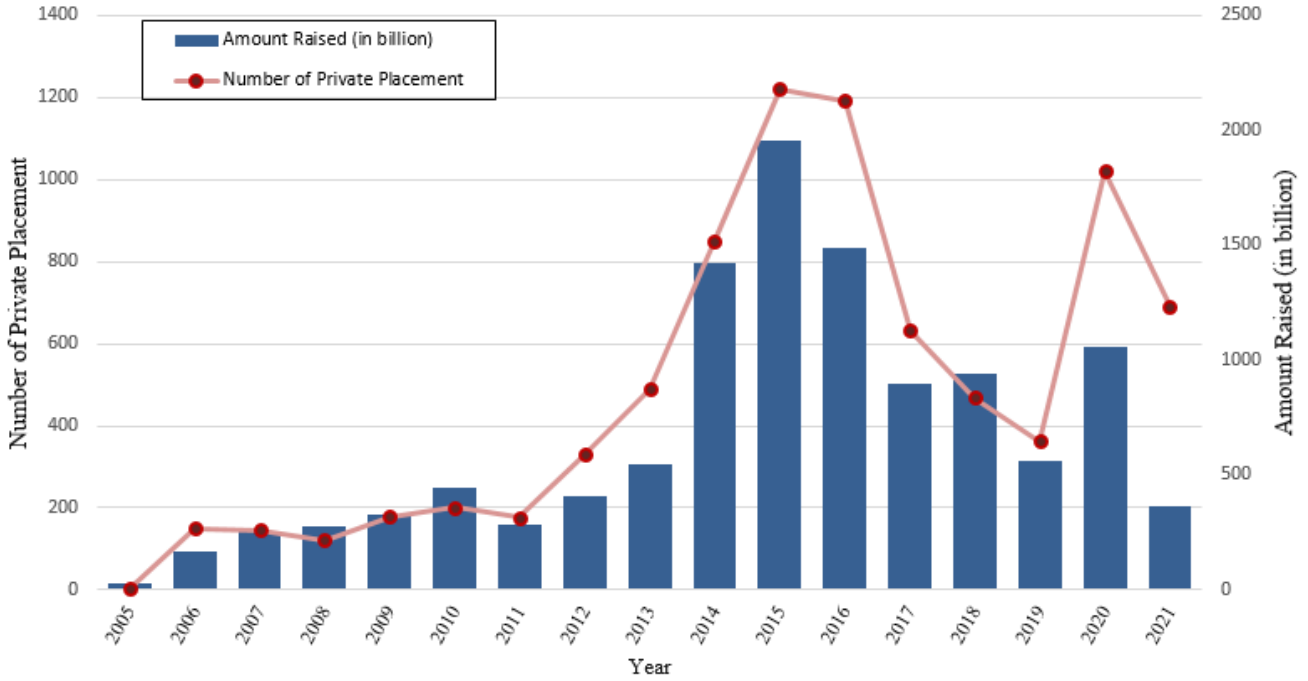
Number of domestic operating companies (CRSP share codes 10 and 11) listed on major exchanges at quarter-end. The counts include SPACs that have not yet announced a merger but have separately traded shares, but exclude SPACs that intend to merge with a "global" target.
Source: CRSP

USA 2022 GDP USD 25 Trillion
Stock Market Cap USD 50 Trillion

Number of IPOs and Amount Raised, 2000 - 2021

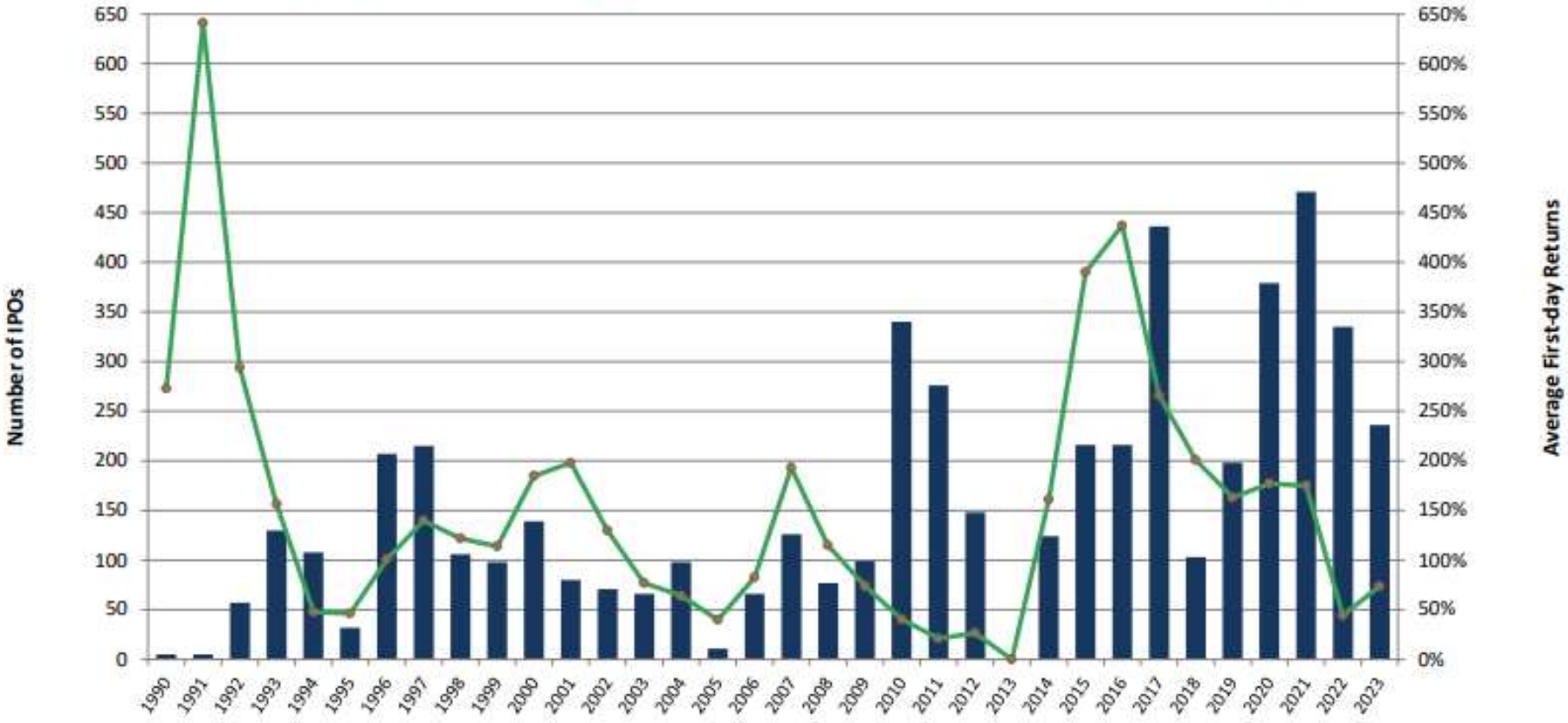


Number of Private Placement and Amount Raised, 2005 - 2021

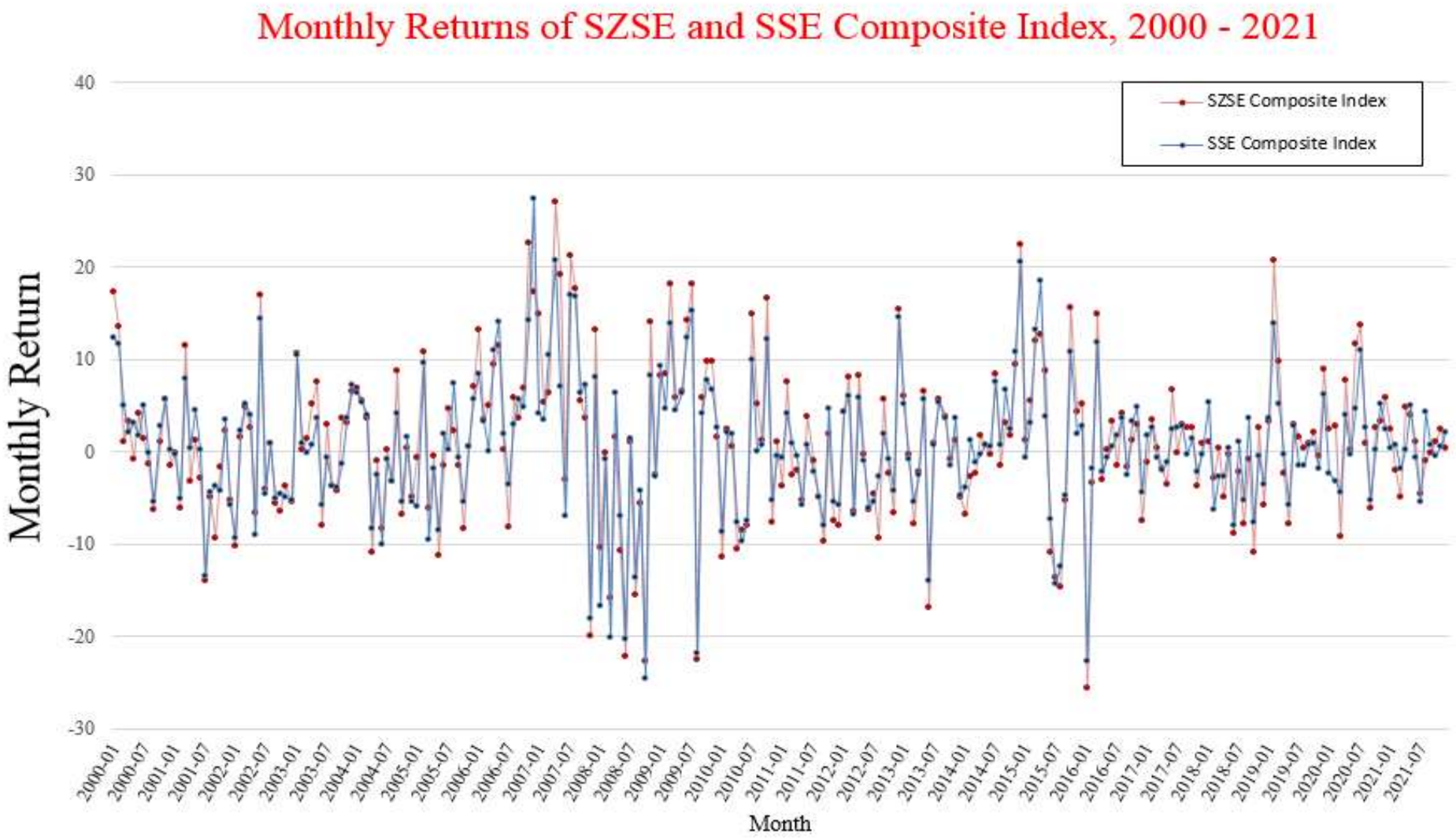


Number of Offerings and Average First-day Returns on Chinese IPOs, 1990-2023

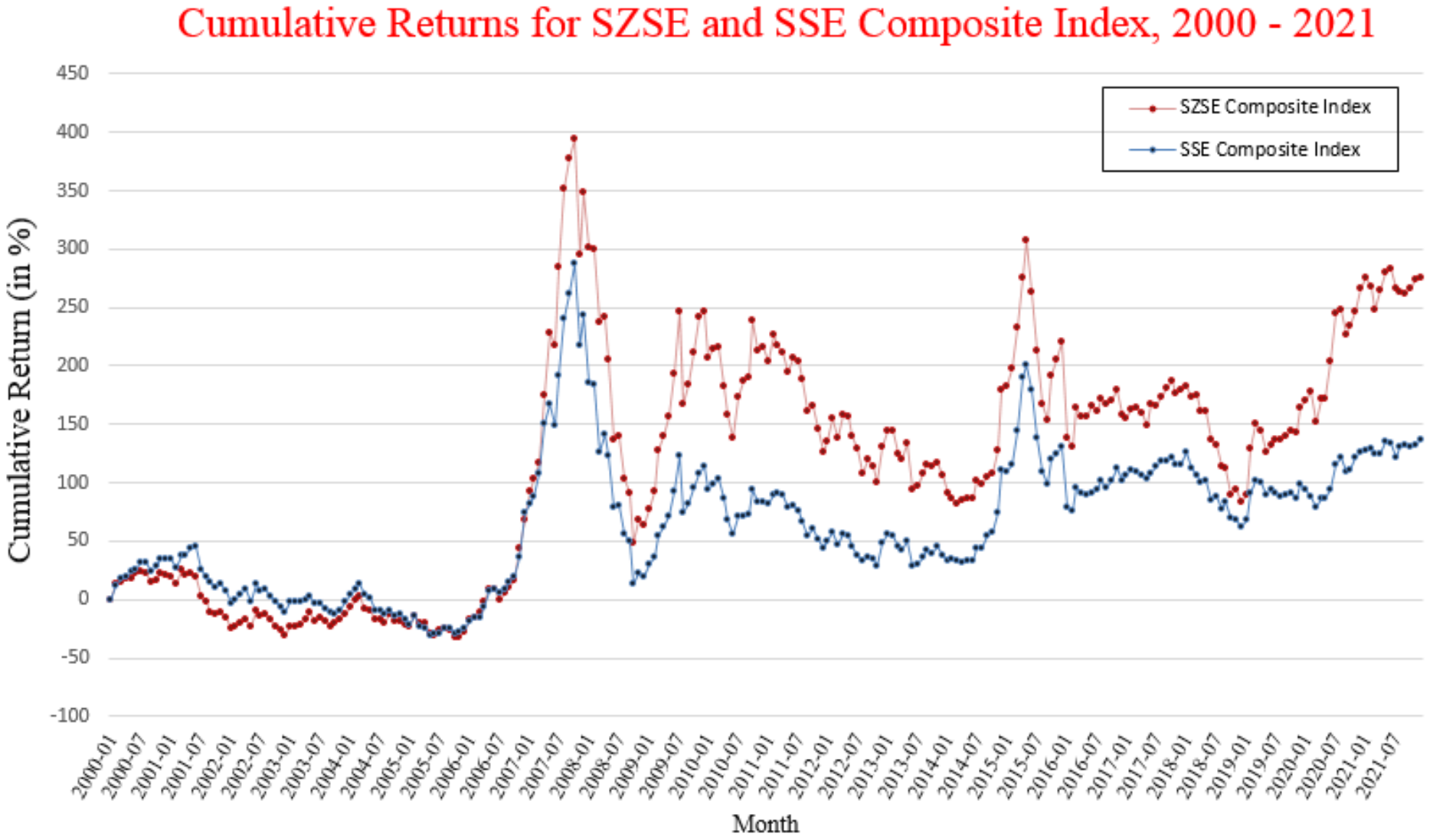
(When price limits are binding, returns are measured until the first close that is unconstrained)

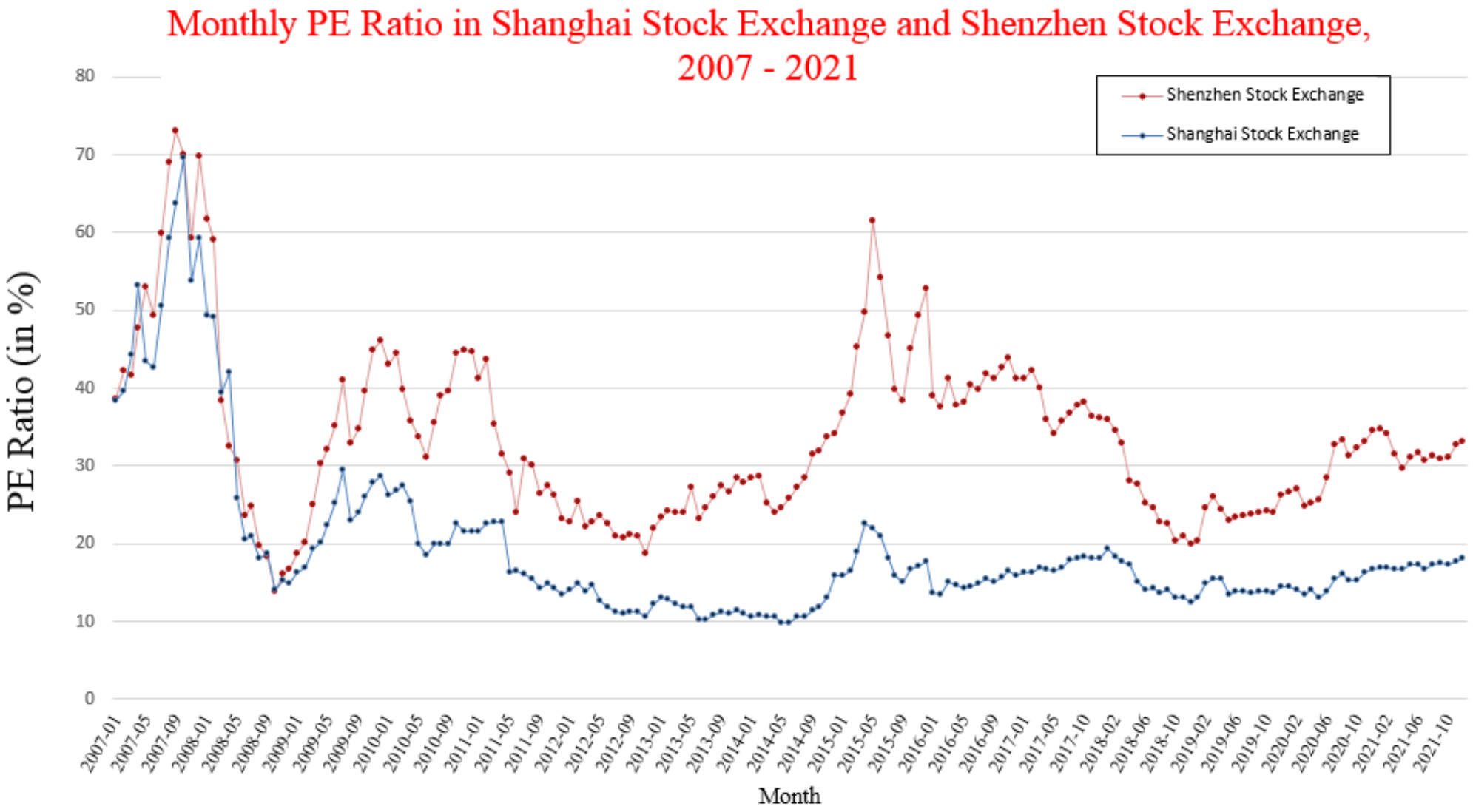


Source: Qian, Ritter, and Shao "Initial Public Offerings Chinese Style" (*JFQA* 2024) and Dealogic. There were no IPOs in 2013, due to a CSRC moratorium.

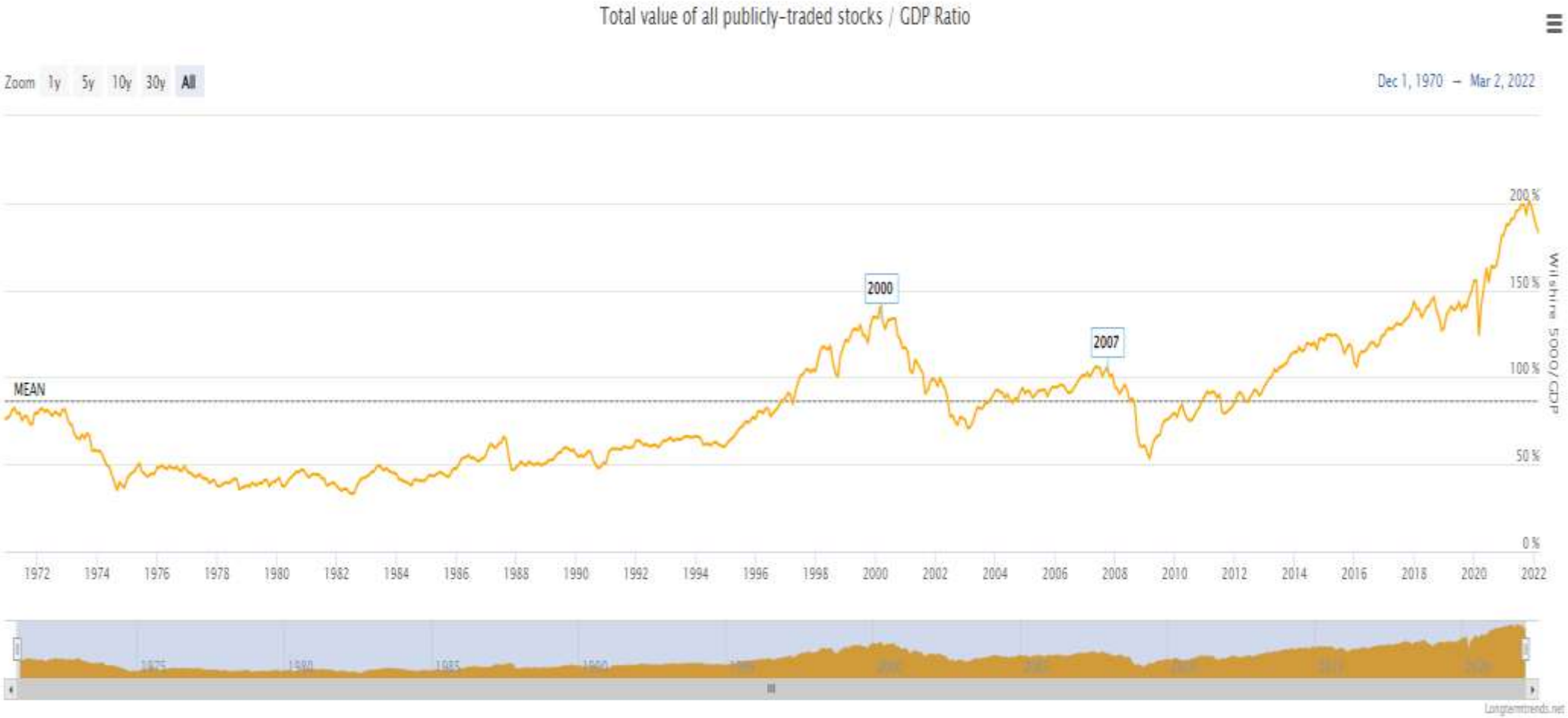


SSE: mean=0.64% std=7.28%
SZSE: mean=0.90%, std=8.27%


Cumulative Returns for SZSE and SSE Composite Index, 2000 - 2021
SZSE Composite Index
SSE Composite Index
Cumulative Return (in %)
450
400
350
300
250
200
150
100
50
0
-50
-100
2000-01
2000-07
2001-01
2001-07
2002-01
2002-07
2003-01
2003-07
2004-01
2004-07
2005-01
2005-07
2006-01
2006-07
2007-01
2007-07
2008-01
2008-07
2009-01
2009-07
2010-01
2010-07
2011-01
2011-07
2012-01
2012-07
2013-01
2013-07
2014-01
2014-07
2015-01
2015-07
2016-01
2016-07
2017-01
2017-07
2018-01
2018-07
2019-01
2019-07
2020-01
2020-07
2021-01
2021-07
Month


Monthly PE Ratio in Shanghai Stock Exchange and Shenzhen Stock Exchange, 2007 - 2021
Shenzhen Stock Exchange
Shanghai Stock Exchange
PE Ratio (in %)
Month
0
10
20
30
40
50
60
70
80
2007-01
2007-05
2007-09
2008-01
2008-05
2008-09
2009-01
2009-05
2009-09
2010-01
2010-05
2010-09
2011-01
2011-05
2011-09
2012-01
2012-05
2012-09
2013-01
2013-05
2013-09
2014-01
2014-05
2014-09
2015-01
2015-05
2015-09
2016-01
2016-05
2016-09
2017-01
2017-05
2017-10
2018-02
2018-06
2018-10
2019-02
2019-06
2019-10
2020-02
2020-06
2020-10
2021-02
2021-06
2021-10

US stock market benchmark



What is a good firm



- High Growth
- High Profitability (high ROE)
- Low Risk (stable growth or cash flow)
- Economic Moat
  - Refers to a business's ability to maintain competitive advantages over its competitors in order to protect its long-term profits and market share.
  - “What we're trying to find is a business that, for one reason or another -- it can be because it's the low-cost producer in some area, it can be because it has a natural franchise because of surface capabilities, it could be because of its position in the consumers' mind, it can be because of a technological advantage, or any kind of reason at all, that it has this moat around it."
- Dividend

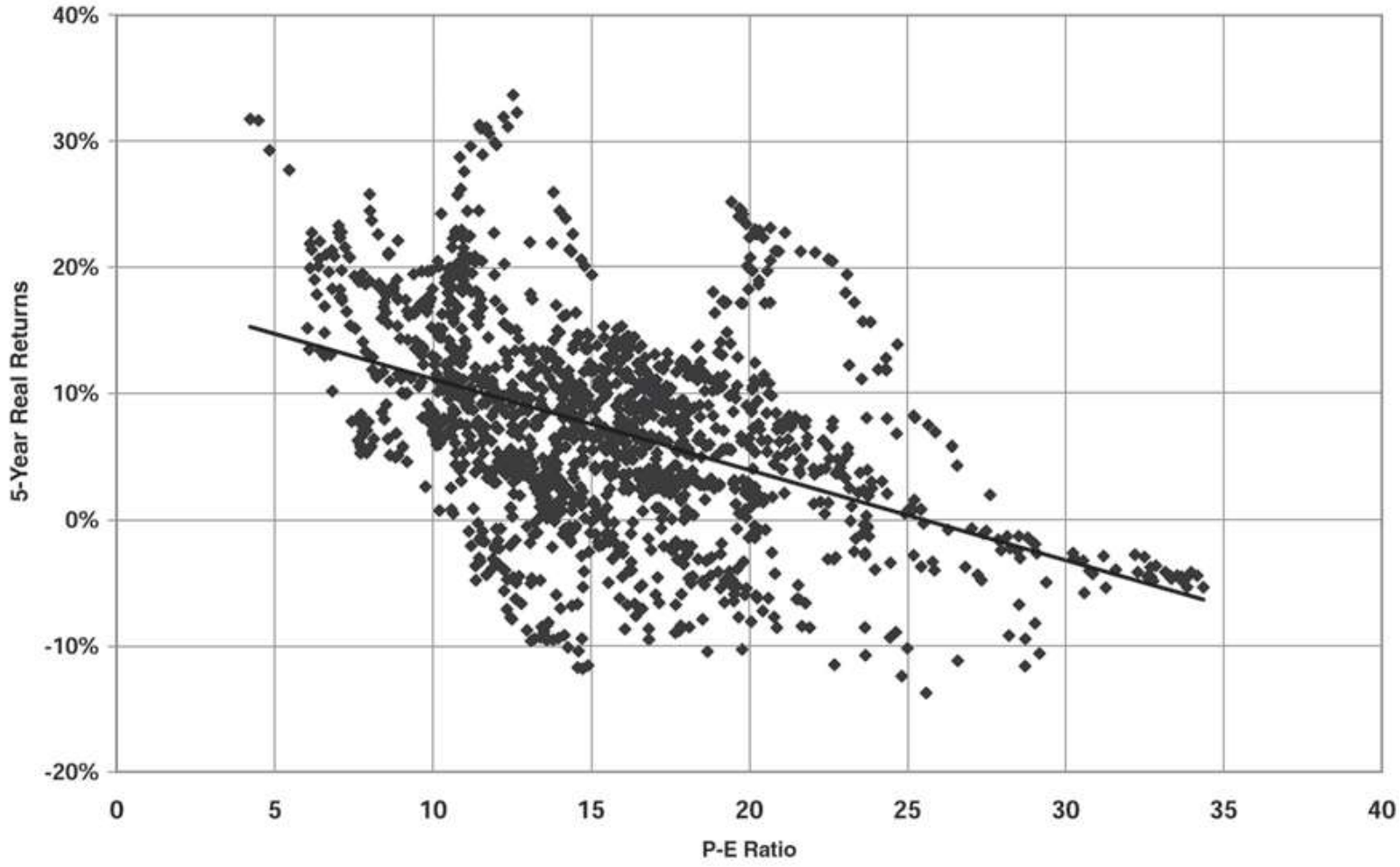
1. "What we're trying to do," he said, answering a question from the audience, "is we're trying to find a business with a wide and long-lasting moat around it, surround -- protecting a terrific economic castle with an honest lord in charge of the castle."

2. "But we are trying to figure out what is keeping -- why is that castle still standing? And what's going to keep it standing or cause it not to be standing five, 10, 20 years from now. What are the key factors? And how permanent are they? How much do they depend on the genius of the lord in the castle?"

3. "And then if we feel good about the moat, then we try to figure out whether, you know, the lord is going to try to take it all for himself, whether he's likely to do something stupid with the proceeds, et cetera.″

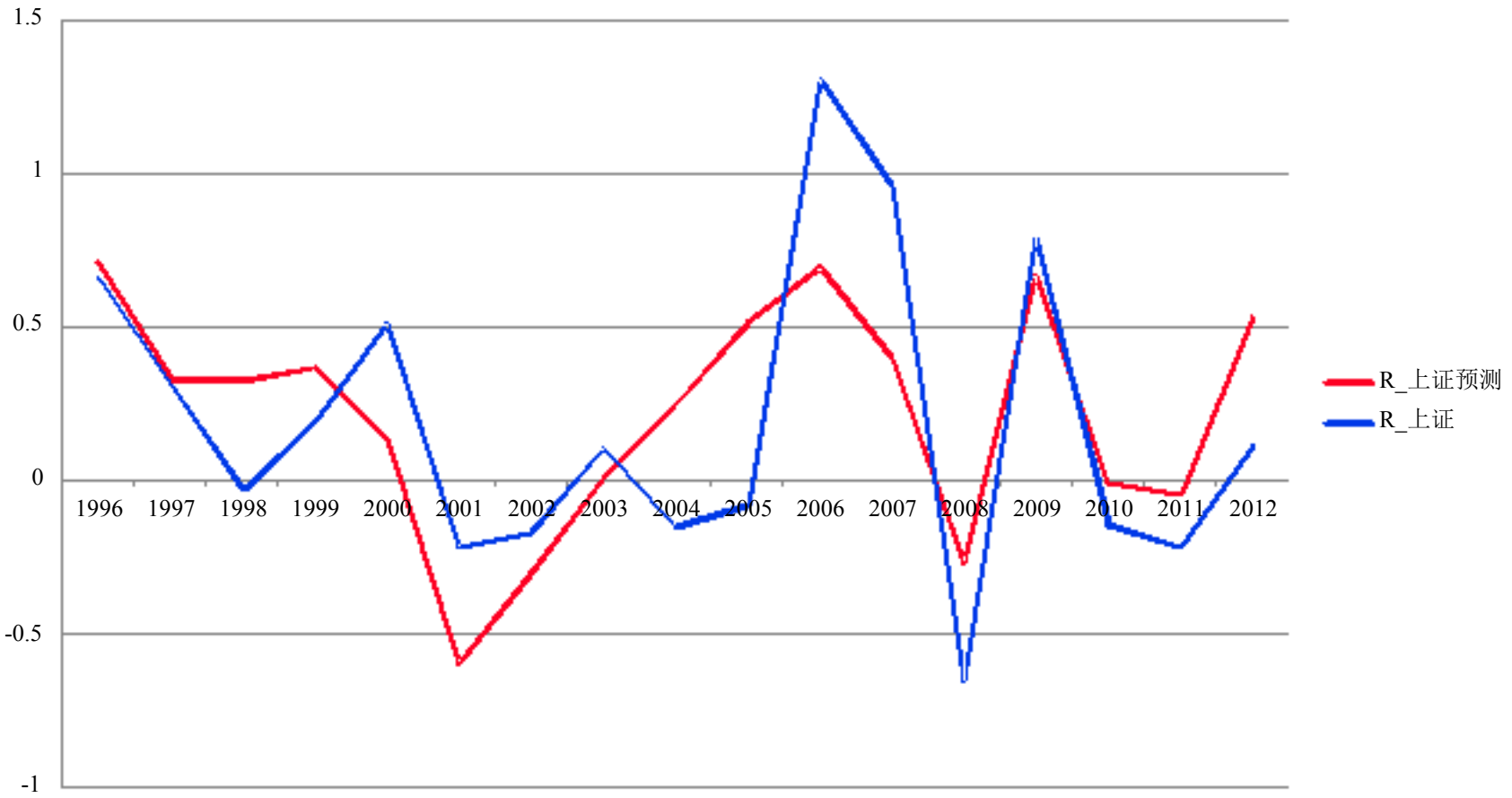
P/E Ratio and the Actual Return of Stock in the next Five Years (Siegel)

Can the return of China stock market be estimated?



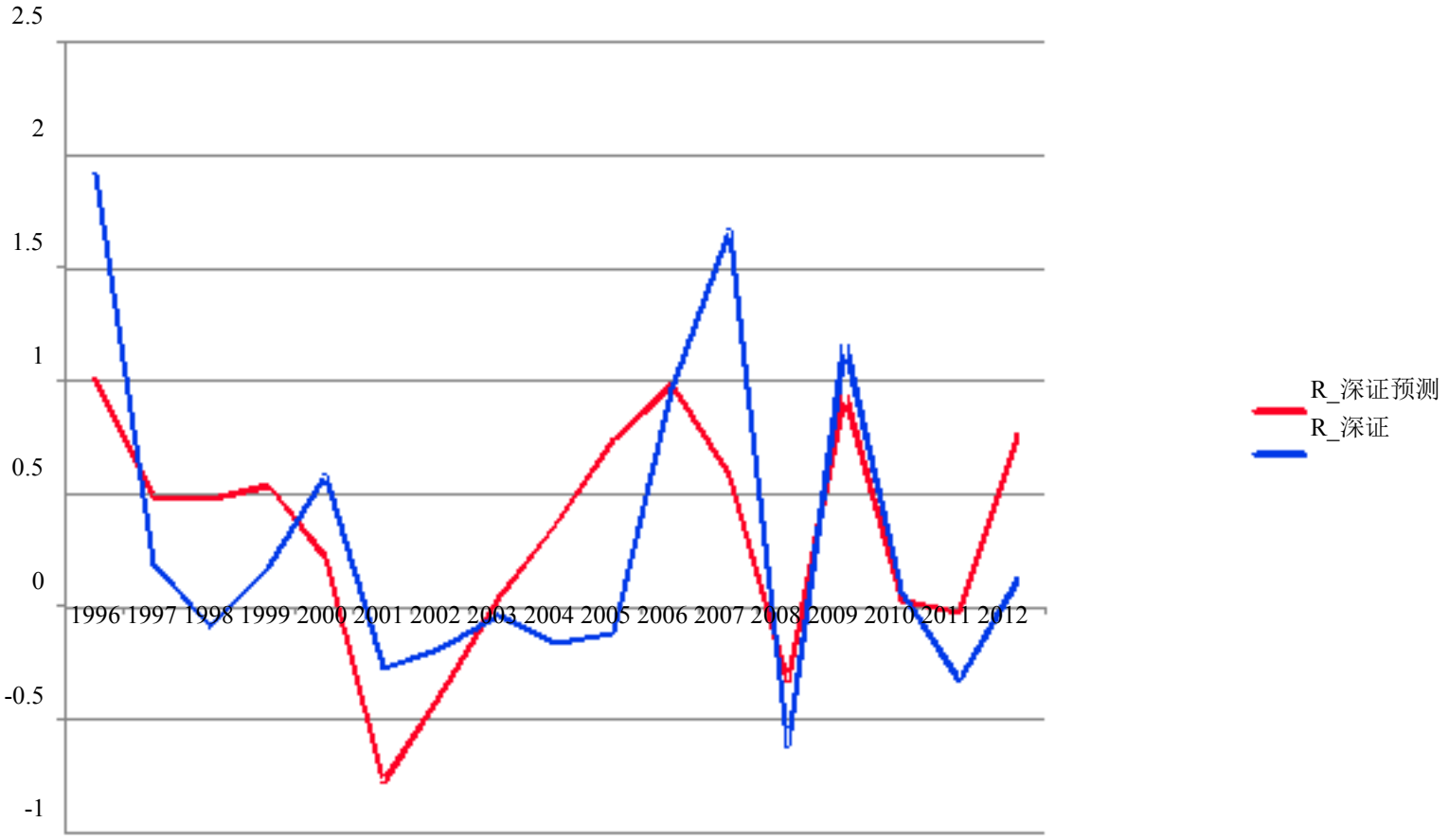
张劲帆

Chinese Market Trend: P/E ratio Estimates Return



张劲帆

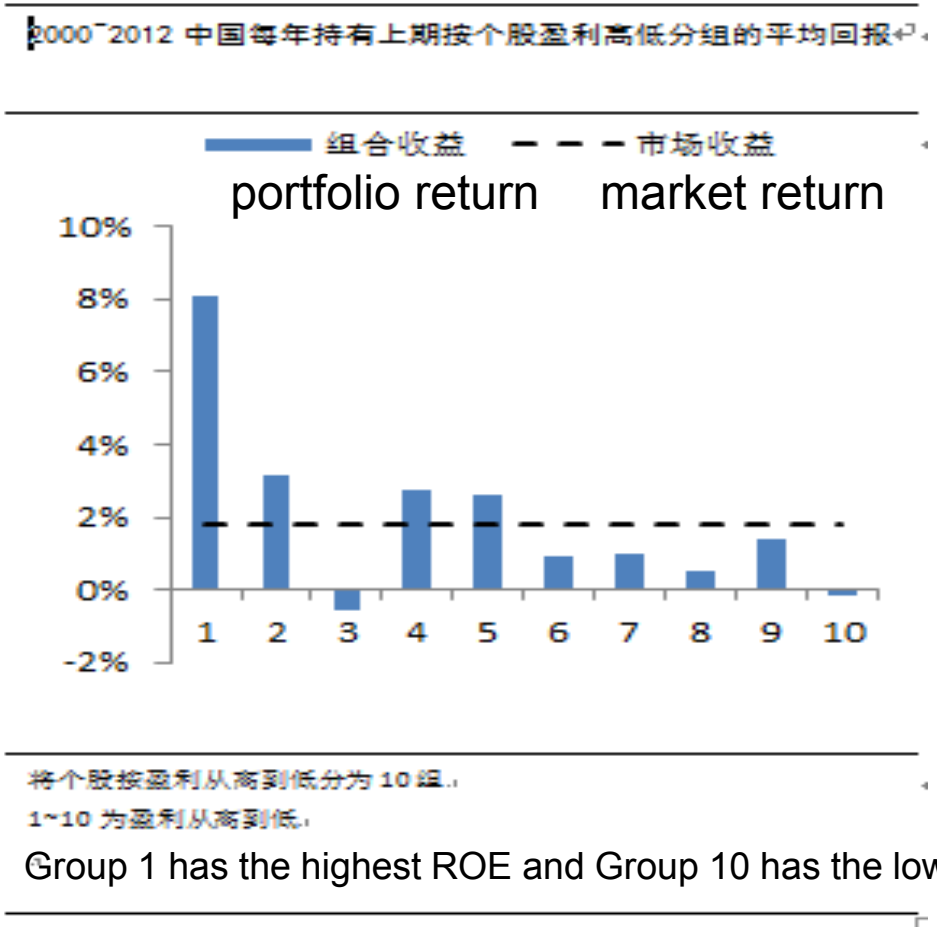
Chinese Market Trend: P/B ratio Estimates Return



张劲帆

Can the Chinese market trend be estimated?

Can the ROE of a stock estimate the future return?



Group 1 has the highest ROE and Group 10 has the lowest ROE

We divide the stocks into 10 groups (1-10) according to the ROE of the stocks at the end of each year . Then, the VW average return is calculated for each group for the following year. We repeat this process every year. Then We calculate the average return for each group.

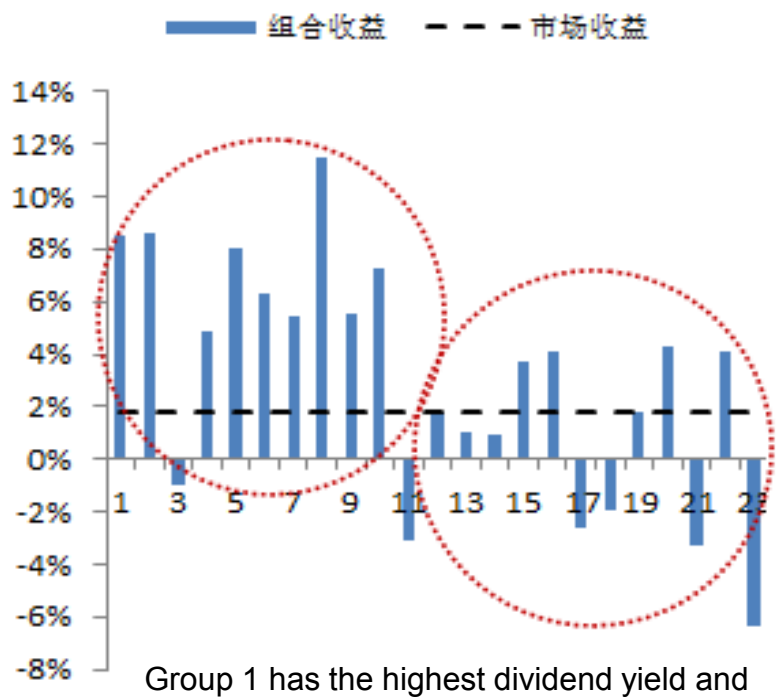
张劲帆

Dividend yield Predicts Return

Can the dividend yield predict the stock return?

At the end of each year, we rank the industries with their average dividend yields and divide them into 23 groups. We then calculate the average return for each group from 2000 to 2012, and compare it with the market. High dividend yield stocks beat the market.

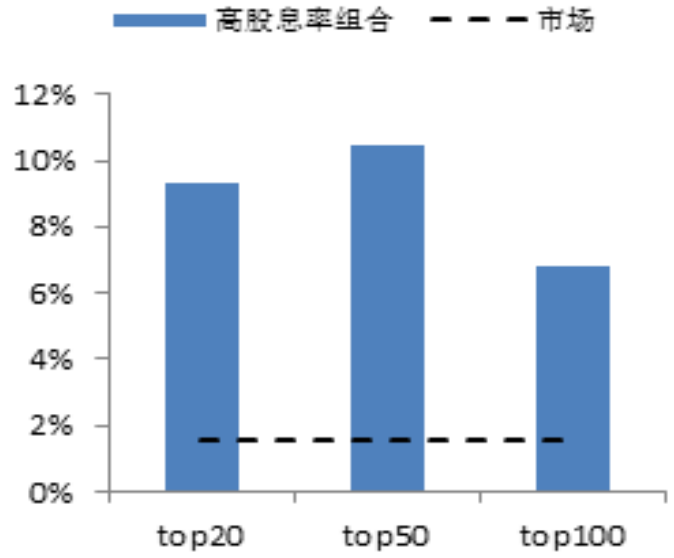
2000~2012 中国每年持有上年按各行业股息率高低排序的平均回报。



Group 1 has the highest dividend yield and Group 23 has the lowest dividend yield

股息率为该表各行业前 5 年平均值后的值；1~23：股息率从高到低；。

2000~2012 中国每年持有上年股息率最高的股票组合取得的回报

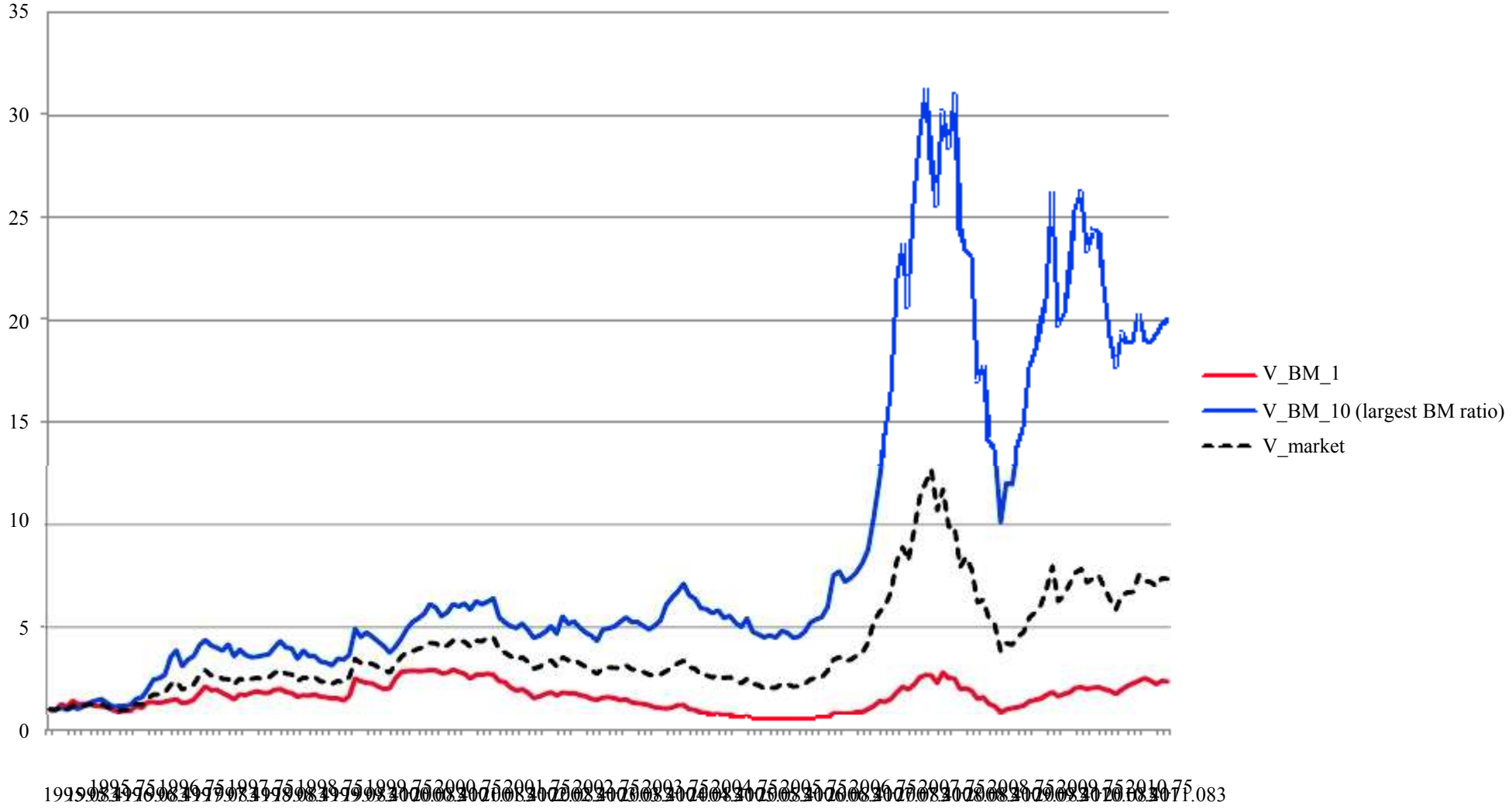


Top20 means the portfolio return of 20 stocks with the highest dividend yield in each year

Top20 为当年股息率最高的 20 只股票，其余以此类推。

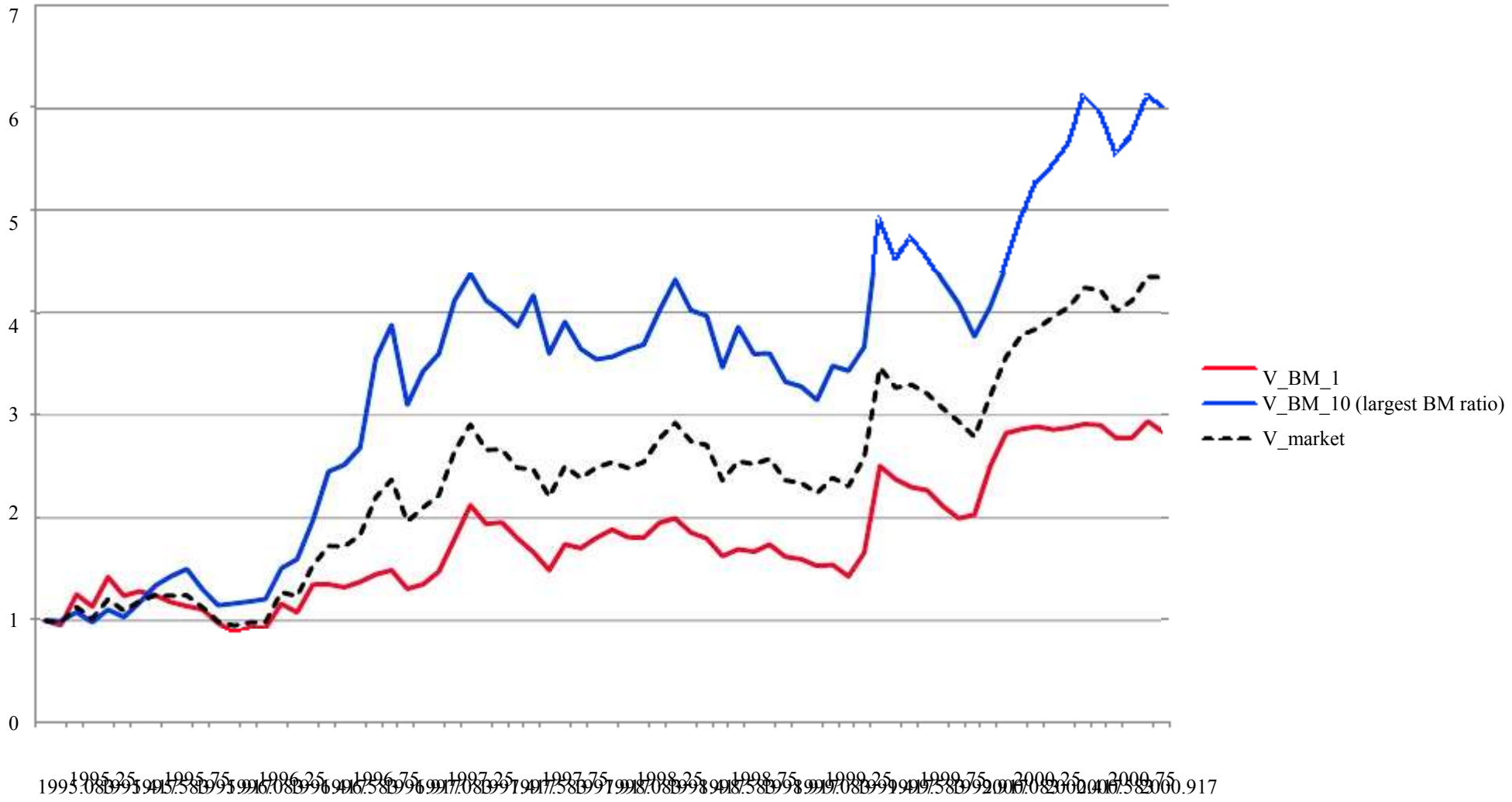
张劲帆

Value Stocks Won the Market: China (1995-2010)



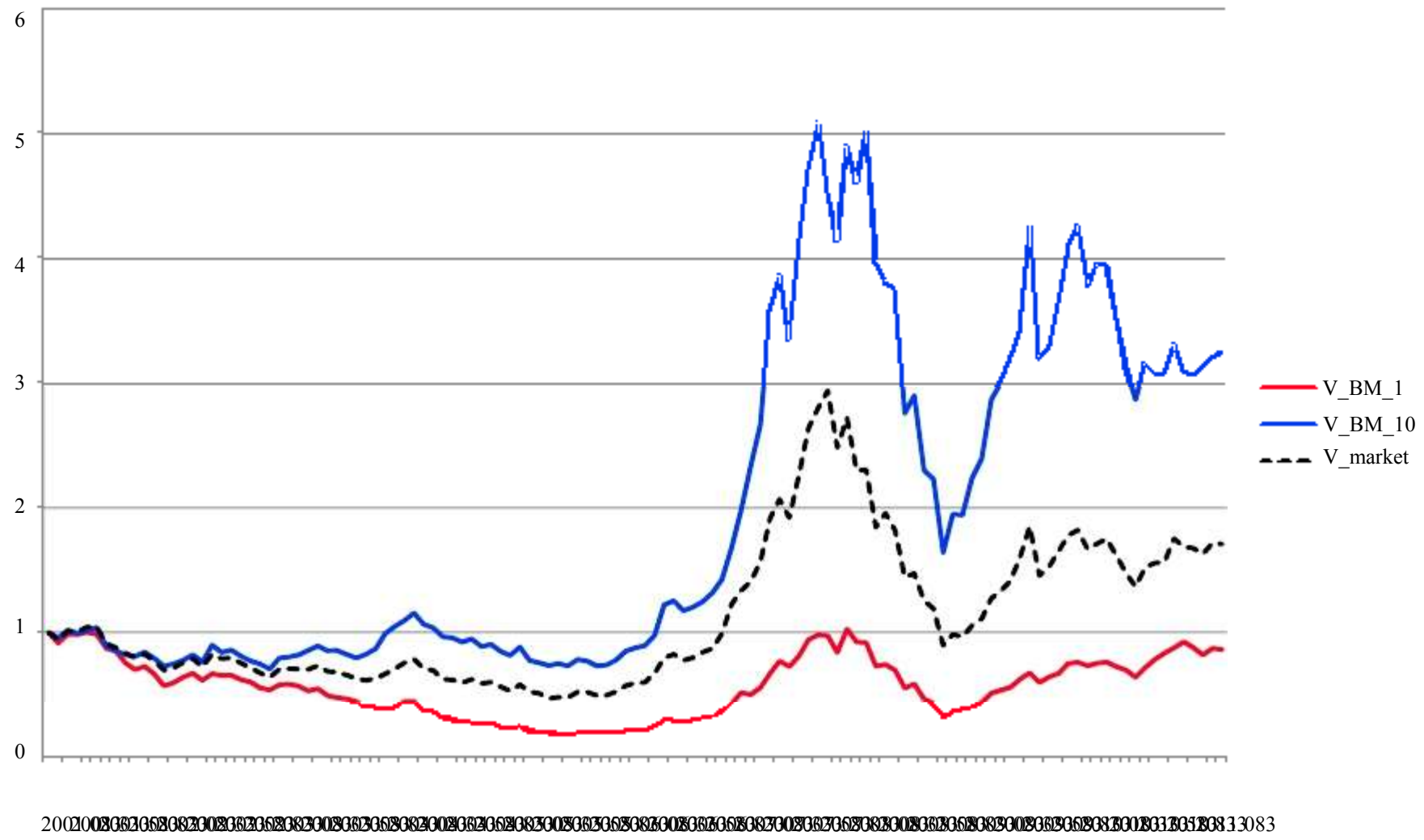
张劲帆

Value Stocks Won the Market: China (1995-2000)



张劲帆

Value Stocks Won the Market: China (2001-2010)



张劲帆

Stock Market Fluctuation and Market Expectation

☆ In the long run, stock must reflect a firm's profitability (the fundamentals of the firm)

☆ If the stock price deviates from its fundamental, it must return

– the larger the price fluctuation, the easier to estimate

– the stock price likes a randomly flying pollen, but in fact subject to gravity and will fall eventually

Long Term Investment and Return

- The relationship between stock market and economy is described by, Andre Kostolany, the German stock market Guru, as “master and his busy dog”. Every master with a dog has similar experiences, in which the dogs often run ahead of you during walking time, and then run back, and sometimes they are left behind. Finally, after arriving at home, the dogs may have run 4 km while the master may only walked 1km. There is a pattern that always holds, the dogs always run around their master.